Introduction to Bioengineering Theory (BBL1020; 3-0-0)

Lecture-11: 2 Classes

Please listen Carefully

&

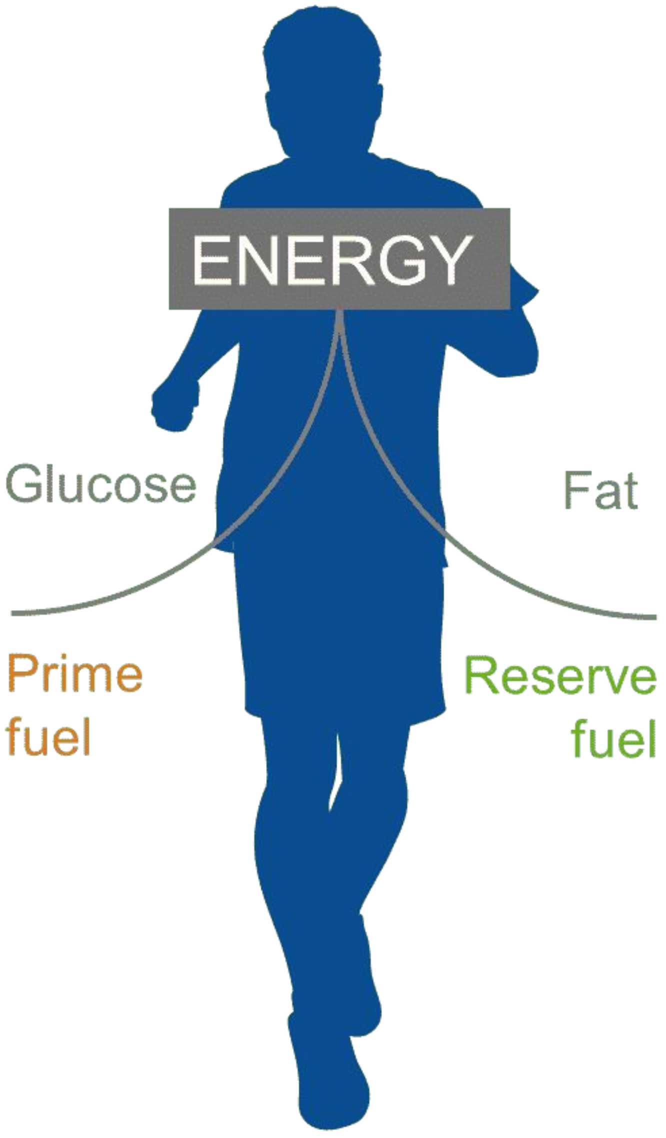
Keep your phone(s) on silent mode

Carbohydrate

- Fast energy supply
- Limited source

Glycogen stores

Depletion of carb fuel limits endurance
(= exhaustion)



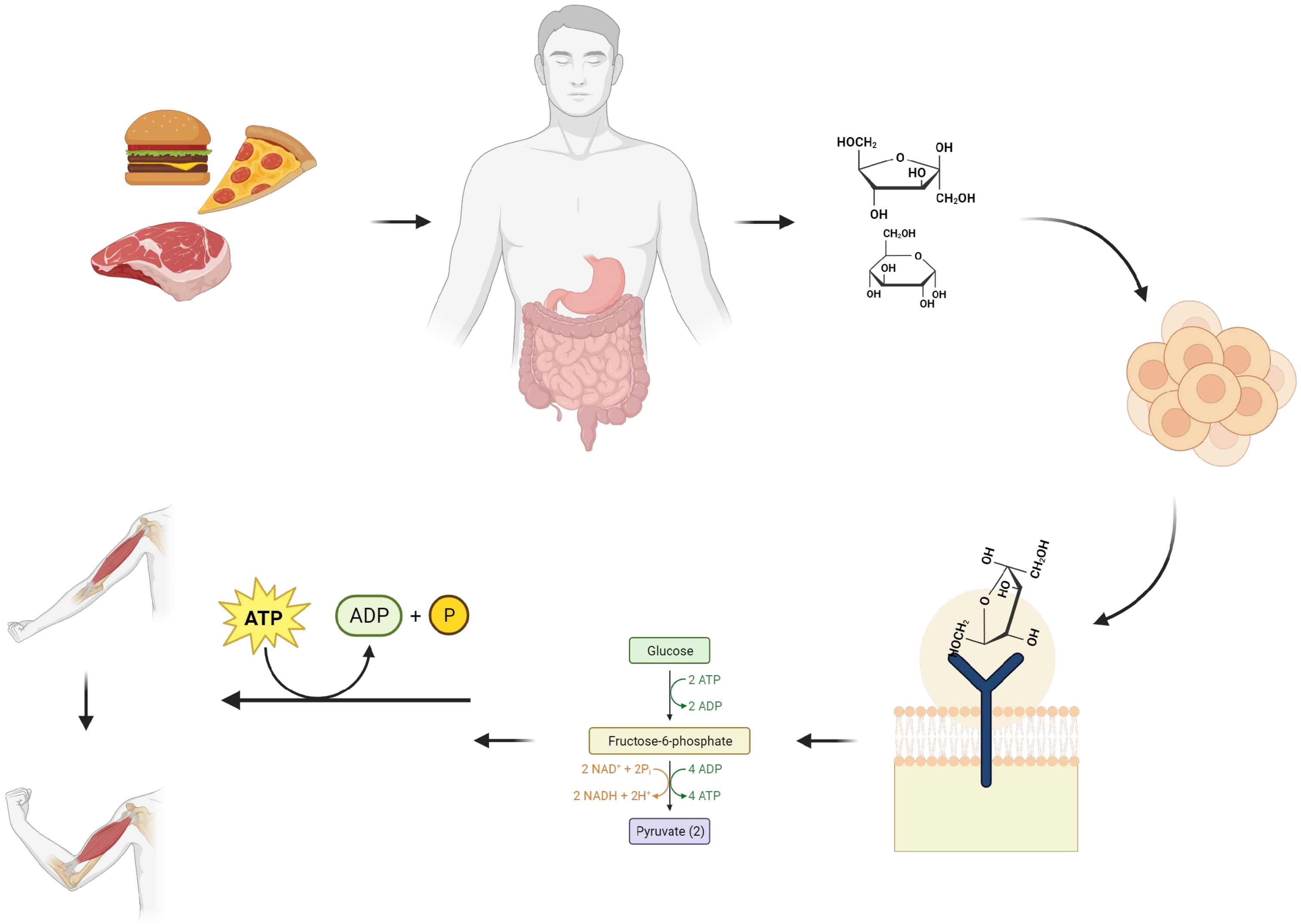
Fat

- Slower energy supply
- Unlimited sources

Fat stores

Carbohydrates

- Hydrates of Carbon.
- General formula- $C_nH_{2n}O_n$
- Aldehydes and ketones with at least two hydroxyl groups.
- Biological evolution selected one form of carbs- D-sugars.
- Building blocks- **Monosaccharides**.



Carbohydrates

Carbohydrates

Monosaccharides (simplest)
-classified according to the no. of carbon

Ex:
~Triose-3C
~Tetrose-4C
~Pentose-5C
~Hexose-6C
~ Heptose-7C
~ Octose-8C
~ Nanose-9C
~Decose-10C

Function
-act as reducing agents when in uncyclized form

Disaccharides
-formed by condensing a pair of monosaccharides units

Ex:
~maltose = glucose + glucose
~Sucrose = glucose + fructose
~Lactose = glucose + galactose

Function
serve a very important role for health and energy

Oligosaccharides
- formed when 2-10 monosaccharides units joined via O-glycosidic bonds

Ex:
~Raffinose = galactose + glucose + fructose

Function
cell recognition & cell binding

Polysaccharides
-contain number of monosaccharides units over 10

Homopolysaccharides
-all one type of monomer

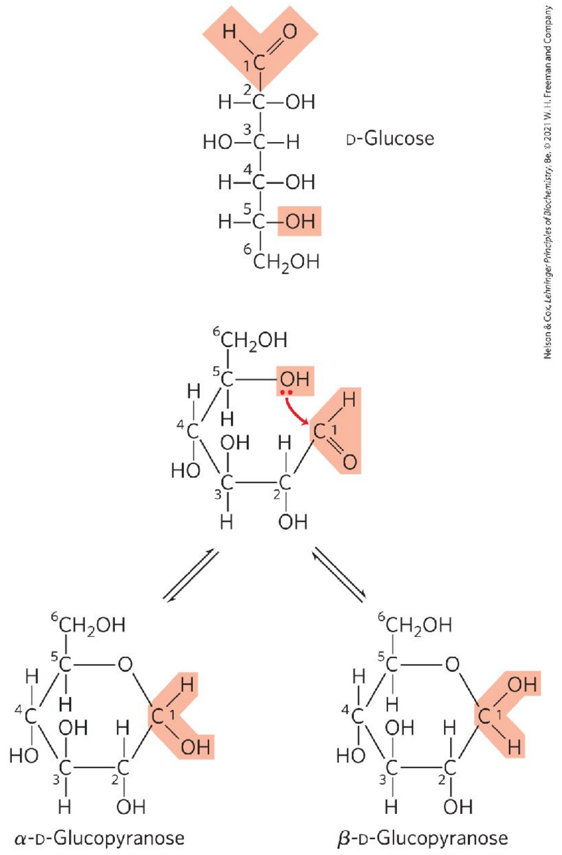
Ex & its function:
~glycogen (storage form in animals)
~starch (storage form in plants)
~cellulose (structure of cell wall)
~chitin (protective cover of insects)

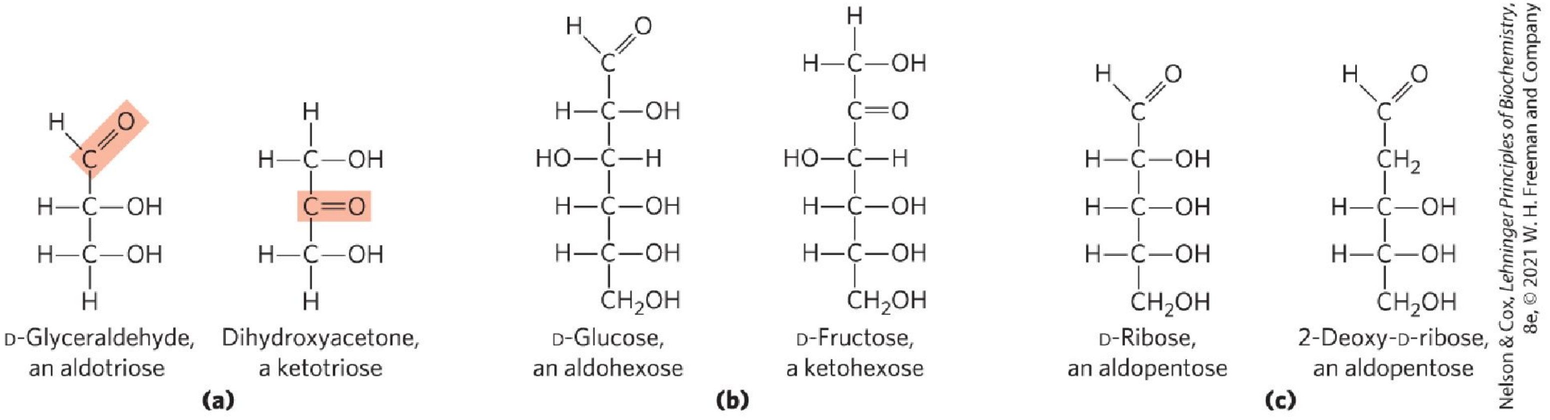
Heteropolysaccharides
-different types of monomers

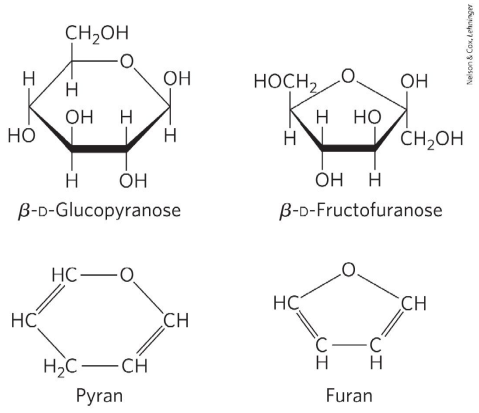
Ex & its function:
~Hemicellulose (strengthening the cell wall)
~peptidoglycans (structural strength to bacterial cell wall)

Monosaccharides

- Simple sugars.
- Single polyhydroxy aldehyde (**aldose**) or ketone (**ketose**) unit.
- Colourless, crystalline solids.
- Freely soluble in water.
- Most have a sweet taste.
- Unbranched C chains in which all Cs are attached by si







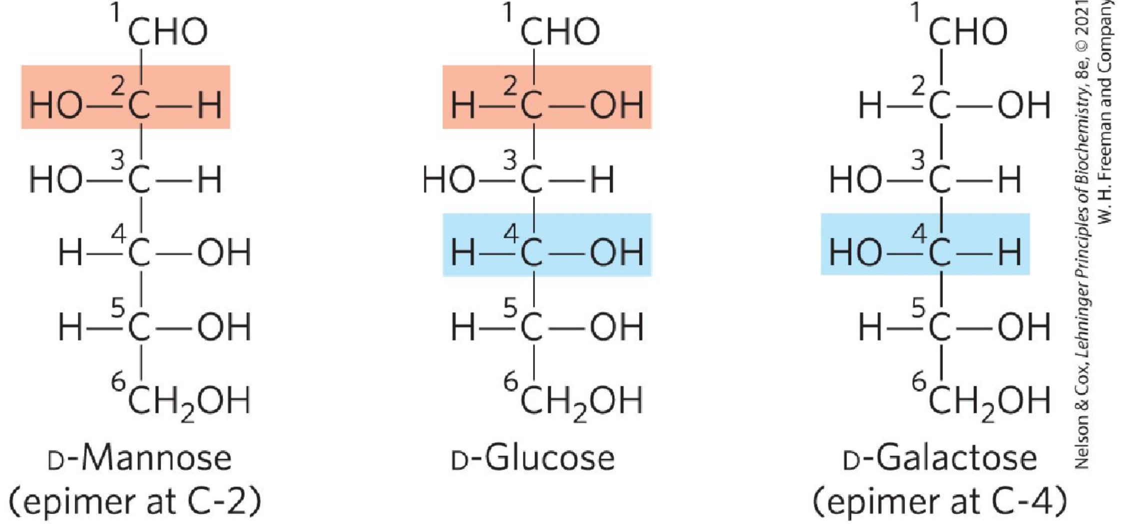
Isomerism in carbs

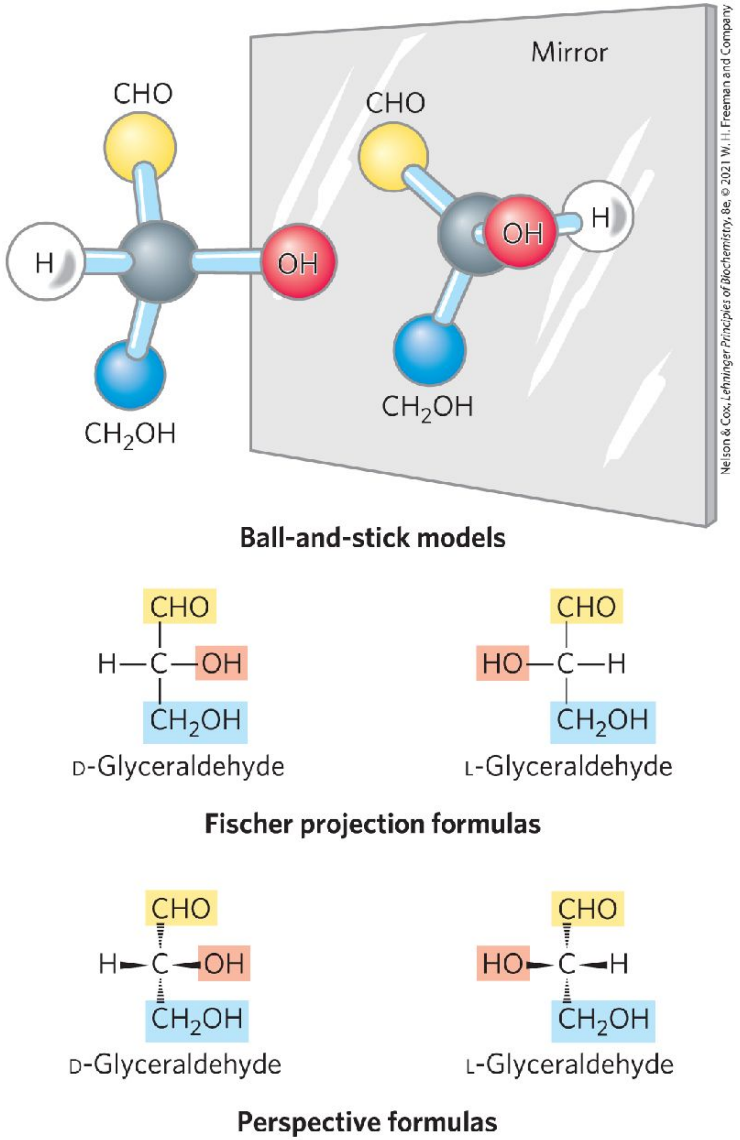
- Most monosaccharides contain one or more chiral C atoms.
- Present as optically active isomeric forms.

- Reference Carbon- Penultimate Carbon.

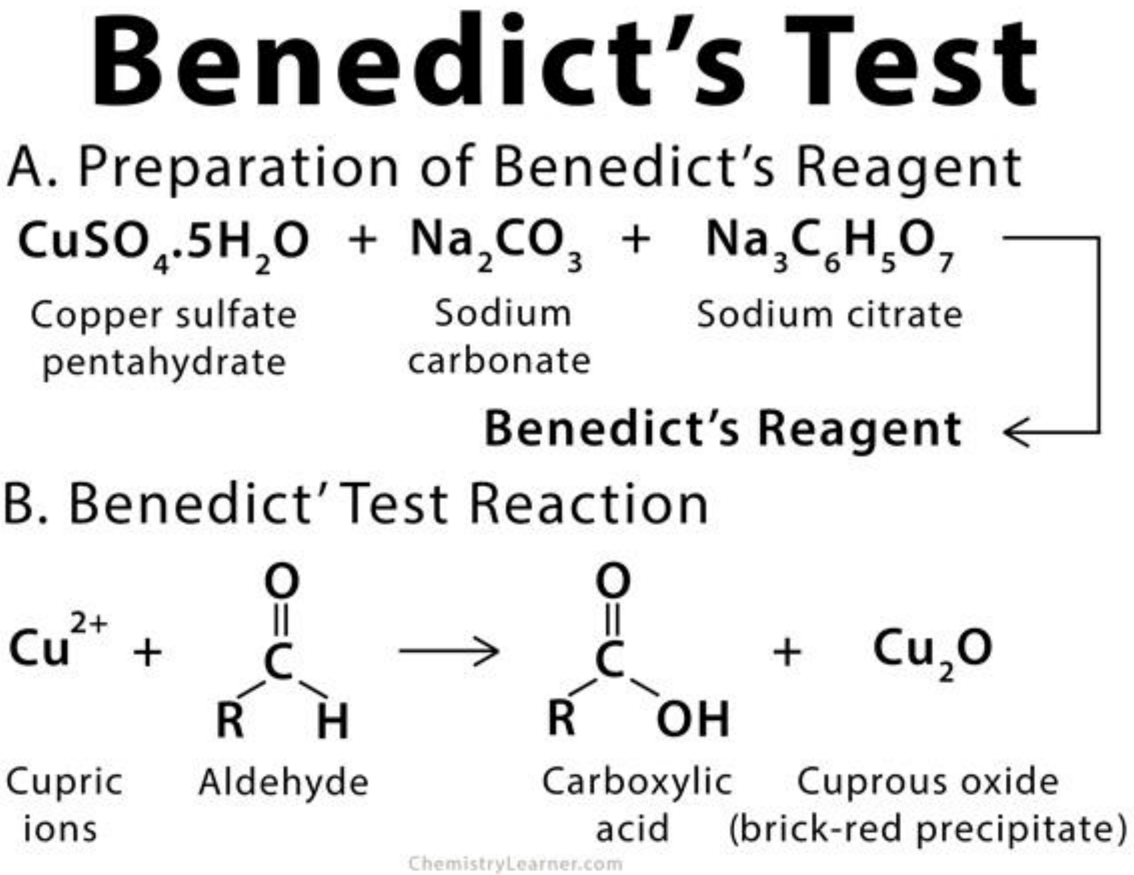
- If OH group is on right side- ***dextro***.
- If OH group is on left side- ***levo***.

- Sugars differing around only 1 C atom- **Epimers.**



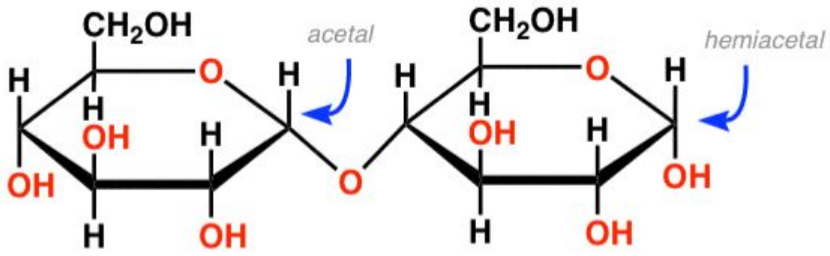


Reducing sugars

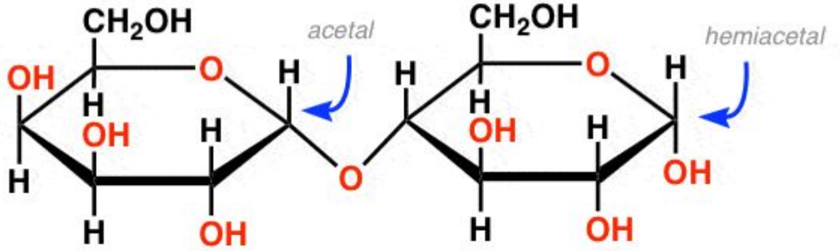


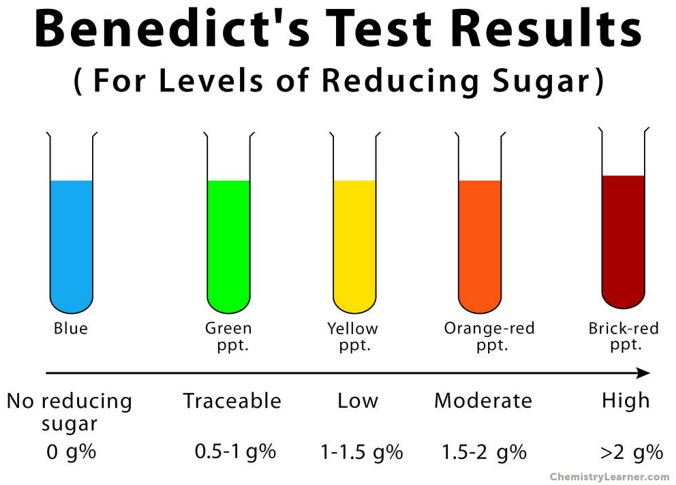
Lactose and Maltose are also reducing sugars and give a positive Benedict test

Maltose:



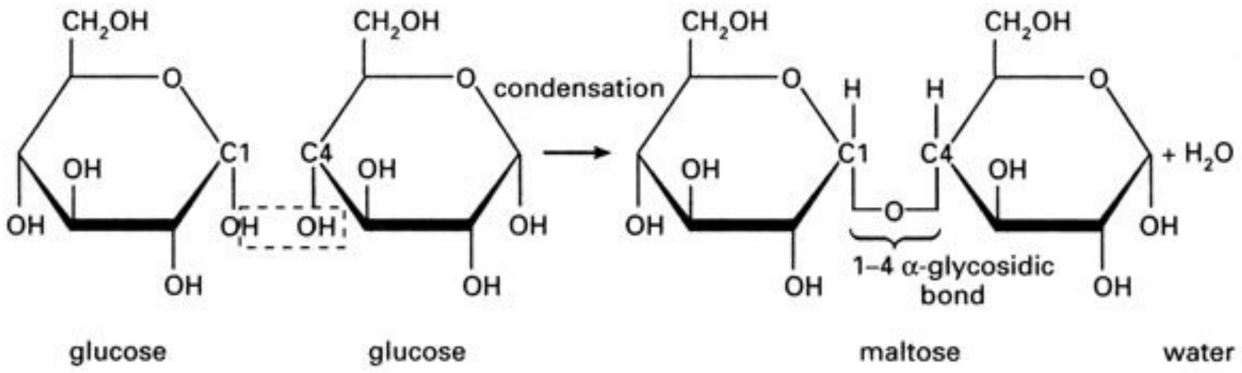
Lactose:

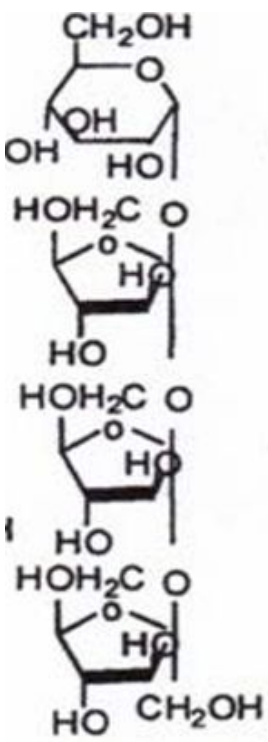


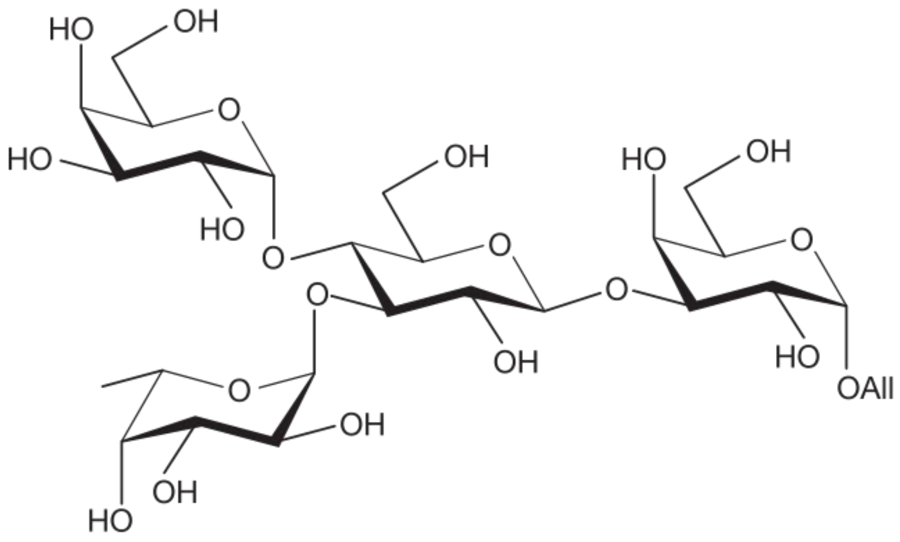


Oligosaccharides

- Carbohydrates composed of a relatively small number of saccharides.
- Formed through glycosidic linkages between hydroxyl groups of two sugar molecules.
- Can be branched or linear depending on the arrangement of sugar units.
- Commonly involved in glycosylation i.e., addition of sugars in biomolecules.





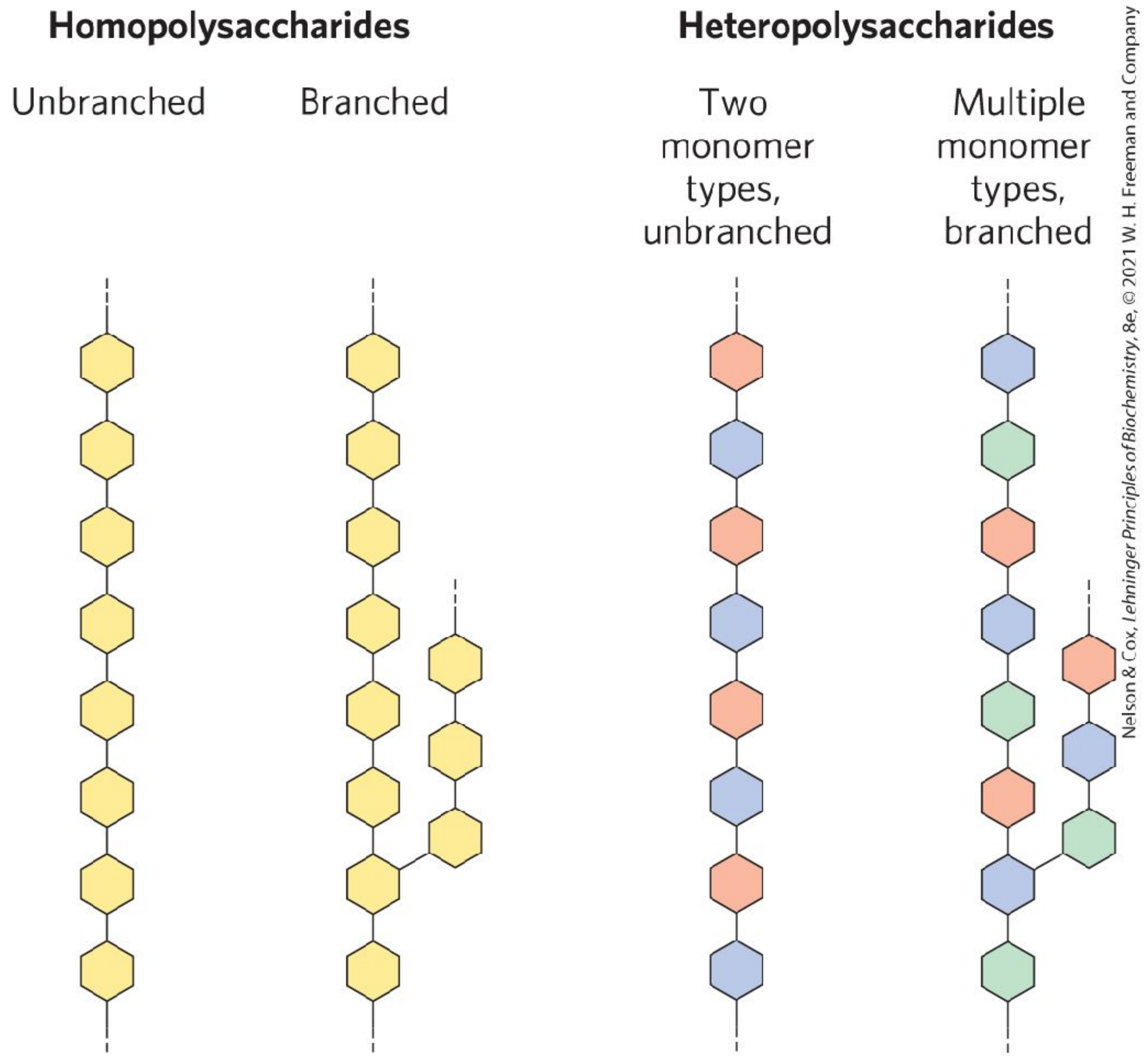


Polysaccharides

- Composed of repeating monosaccharide units.
- Structure of a polysaccharide is intimately linked to its function-

- structural polysaccharides like cellulose form rigid fibers

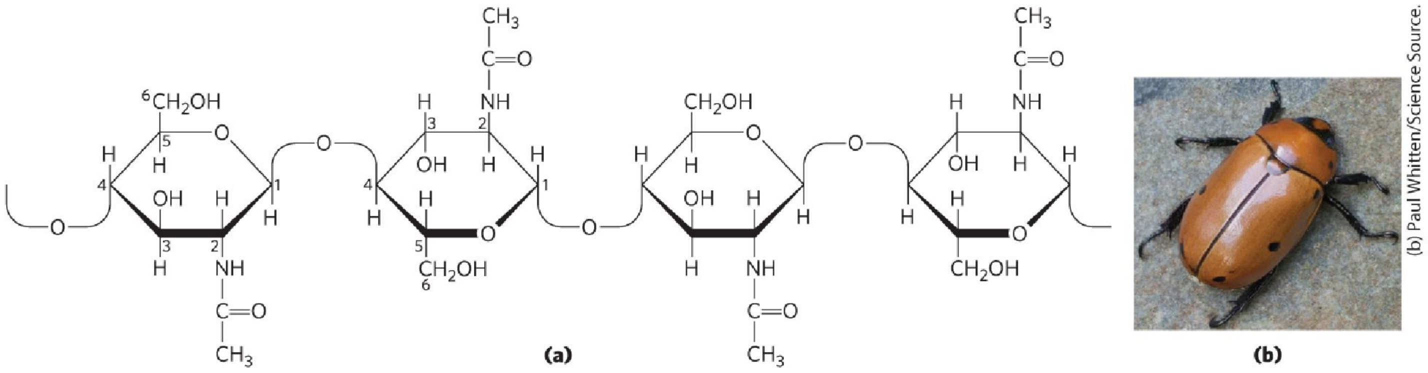
- storage polysaccharides like starch have a more readily accessible structure for enzymatic breakdown.



Polysaccharides

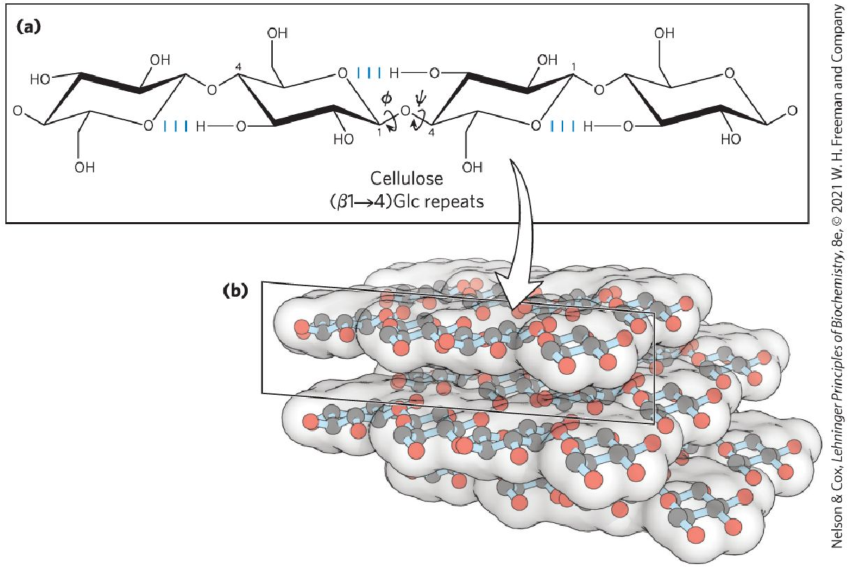
Structural Polysaccharides-

1. Chitin-



- Present in exoskeletons of arthropods.
- Contains N-acetylglucosamine units linked by β-1,4-glycosidic bonds.
- Provides structural support to exoskeleton.

2. Cellulose-



- Found in cell walls of plants.
- Provides structural support and rigidity to cells.
- Contains glucose units linked by β-1,4-glycosidic bonds.

Polysaccharides

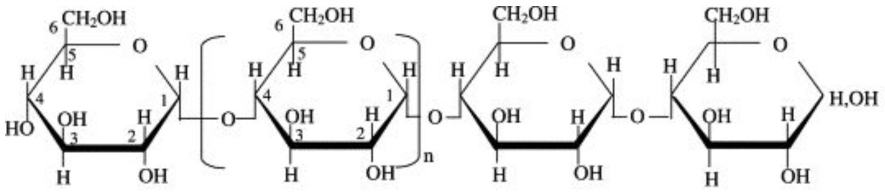
Storage Polysaccharides-

1. **Starch-**

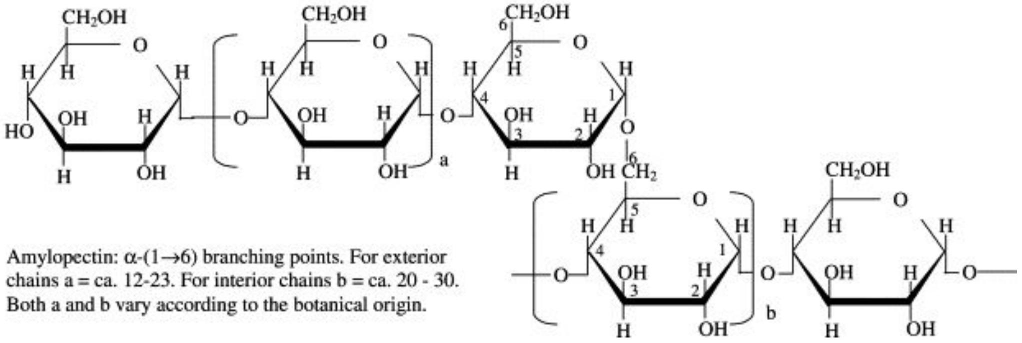
- Found in plants.
- Acts to store energy.
- Consists of two forms- **amylose (α-1,4)** and **amylopectin (α-1,6).**

2. **Glycogen-**

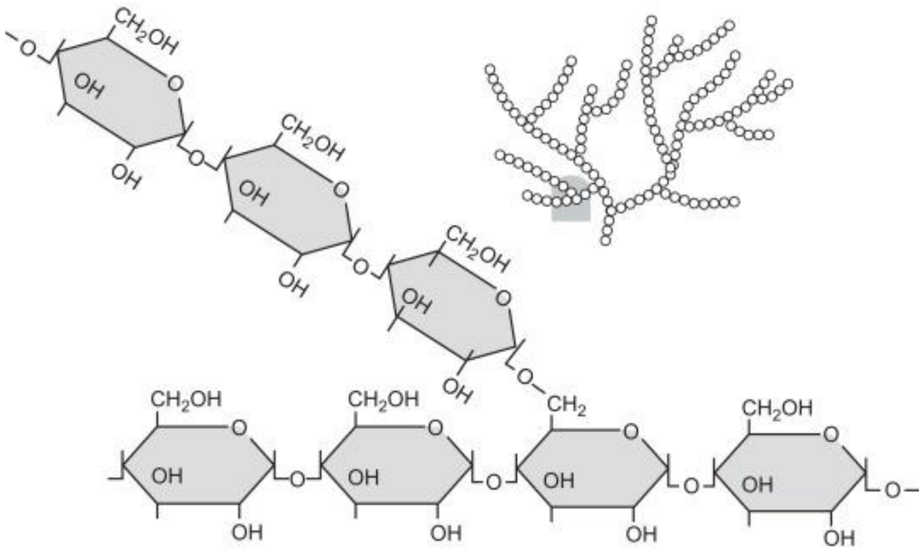
- Found in muscles and liver of animals.
- Helps in storing energy.
- Contains glucose units linked by α-1,4-glycosidic bonds.



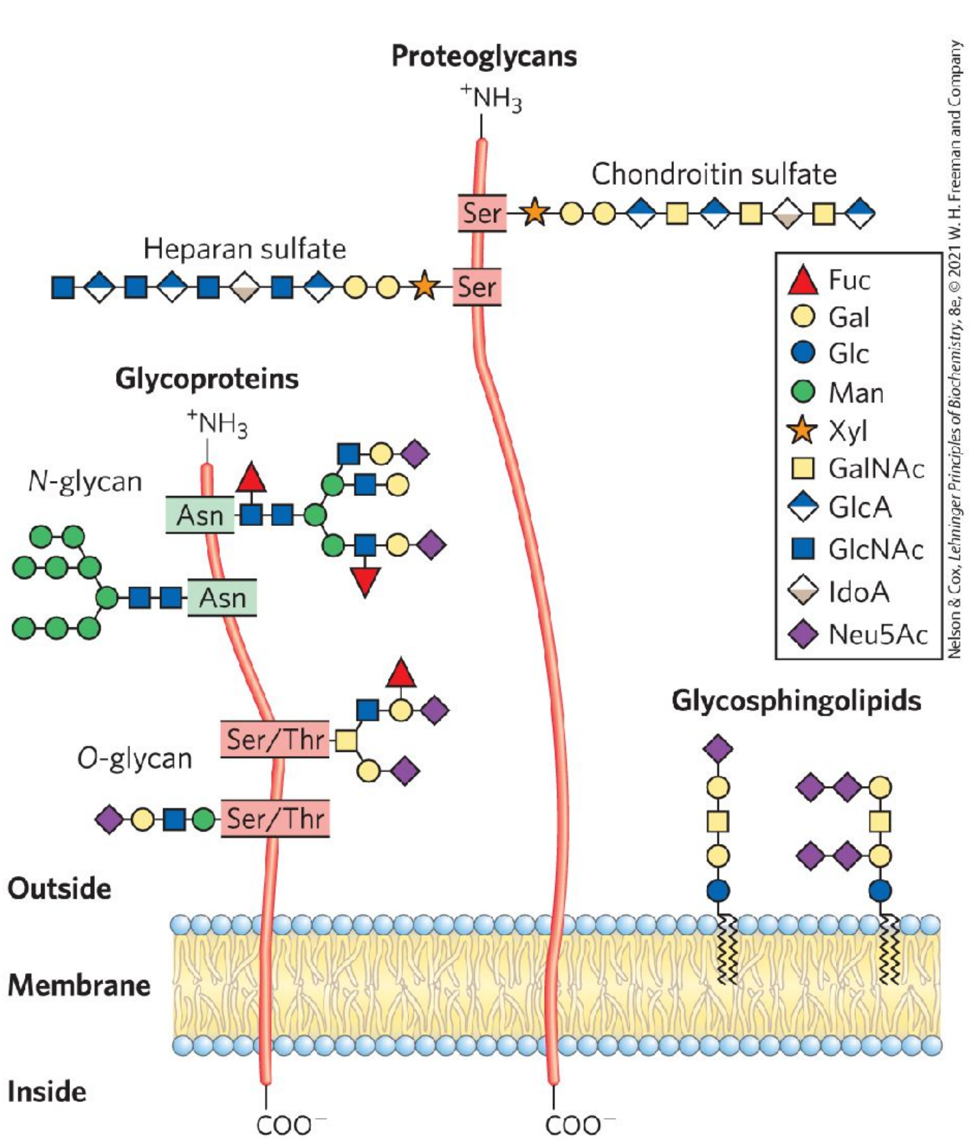
Amylose: α-(1→4)-glucan; average n = ca. 1000. The linear molecule may carry a few occasional moderately long chains linked α-(1→6).

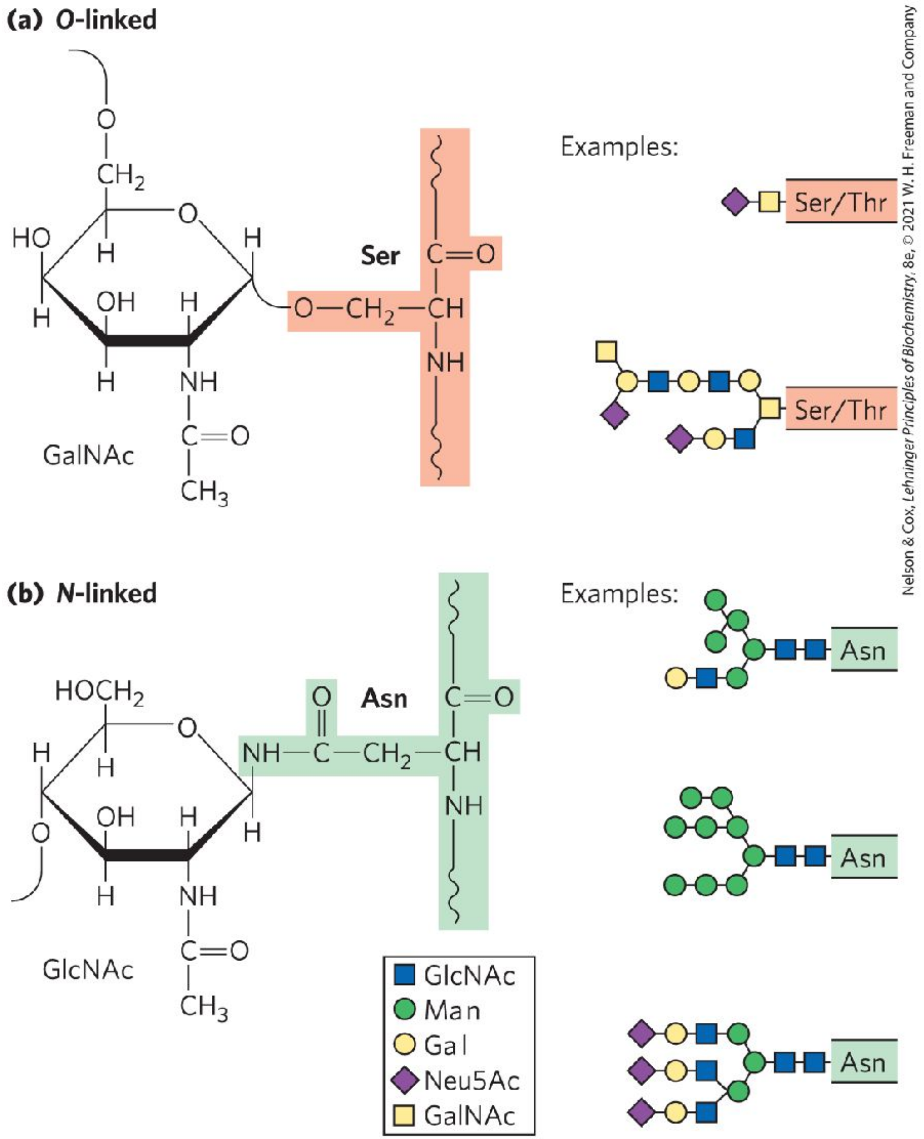


Amylopectin: α-(1→6) branching points. For exterior chains a = ca. 12-23. For interior chains b = ca. 20 - 30. Both a and b vary according to the botanical origin.

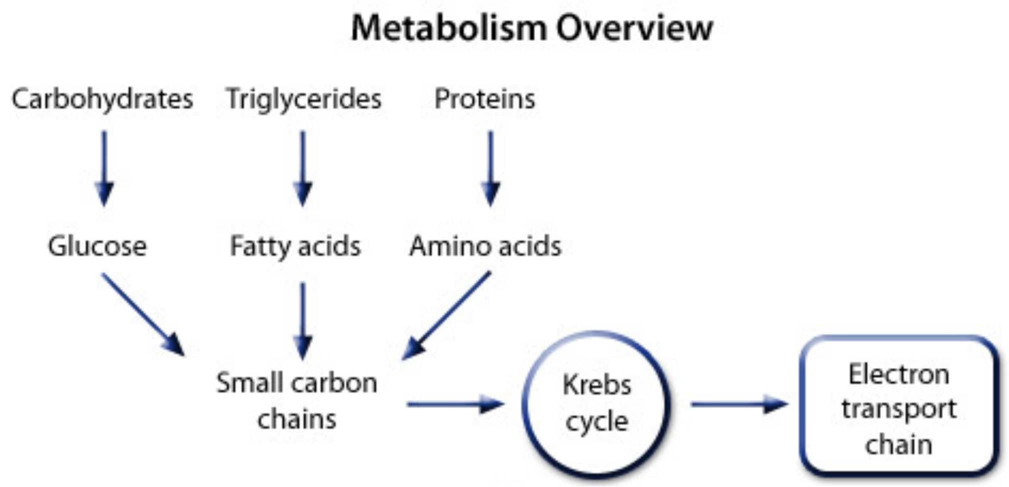


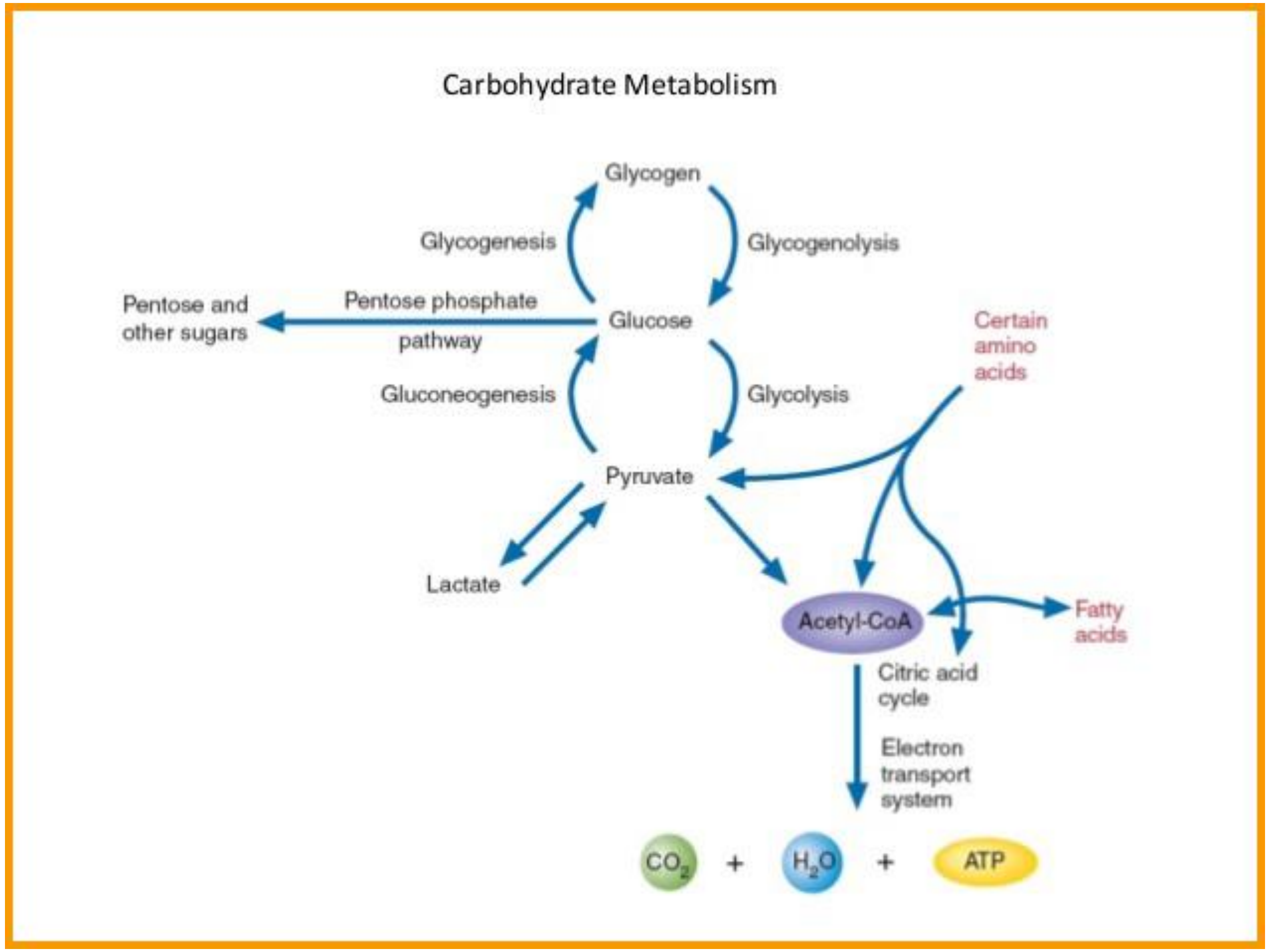
Carbohydrates





Carbohydrates Metabolism





Bio-inspiration

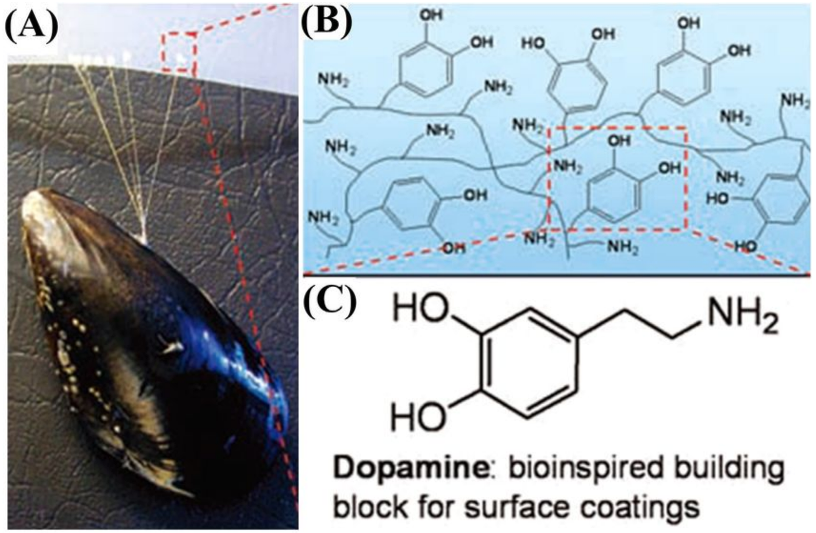
ELSEVIER

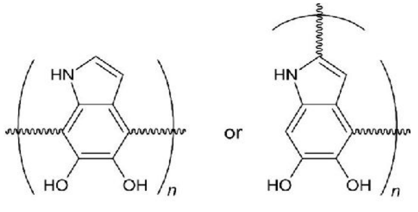
Carbohydrate Polymers
Volume 245, 1 October 2020, 116585

Construction of functional curdlan hydrogels with bio-inspired polydopamine for synergistic periodontal antibacterial therapeutics

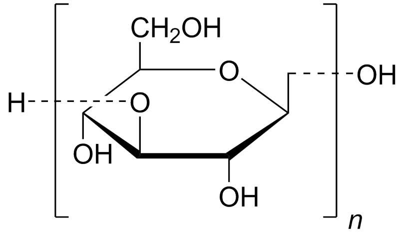
Xiangin Tong [a c 1], Xiaoliang Qi [b c 1], Ruiting Mao [a c 1], Wenhao Pan [a], Mengying Zhang [c], Xuan Wu [c], Gang Chen [c], Jianliang Shen [b c], Hui Deng [a], Rongdang Hu [a]

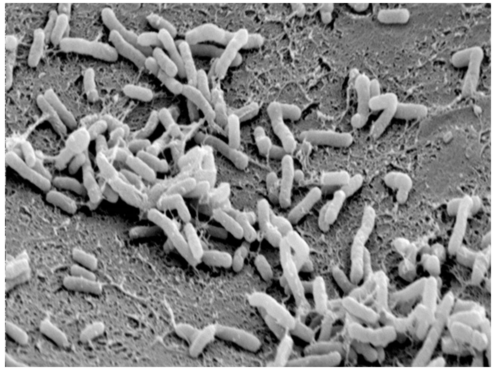
- Curdlan, a bacteria derived polysaccharide with potential for periodontal (gum related infections) antimicrobial delivery.
- PDA – biomimetic self-adherent polymer with NIR photothermal effect.
- CHX – chlorhexidine, a model antimicrobial drug.

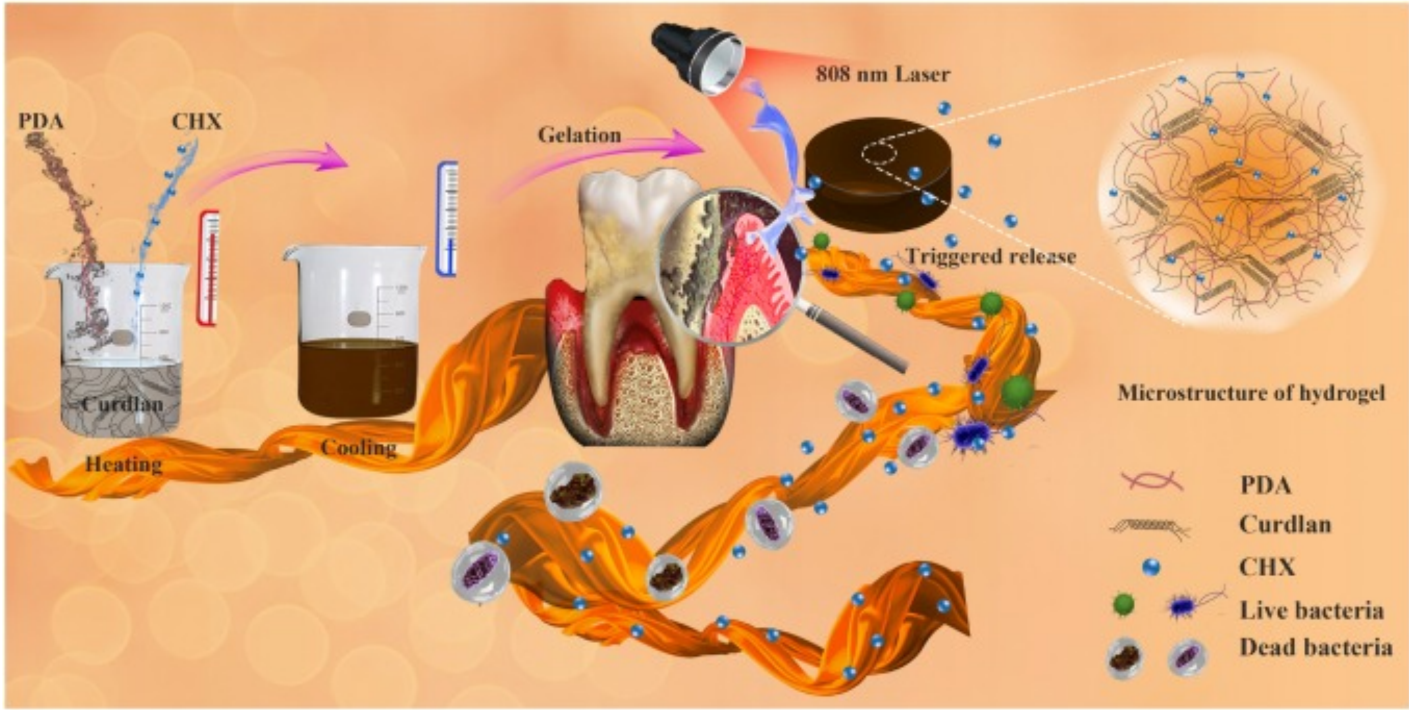




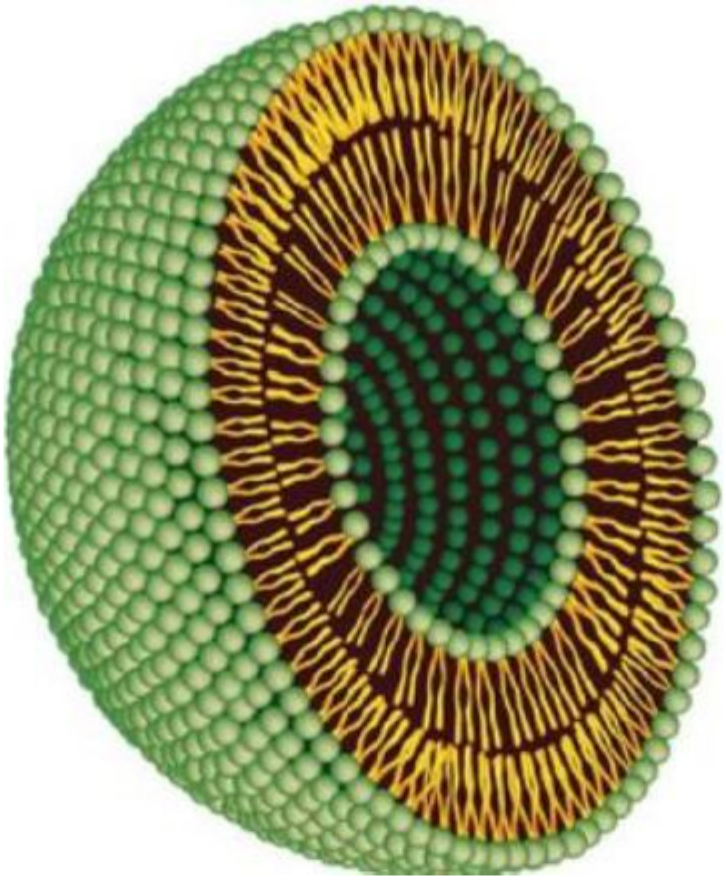
Polydopamine

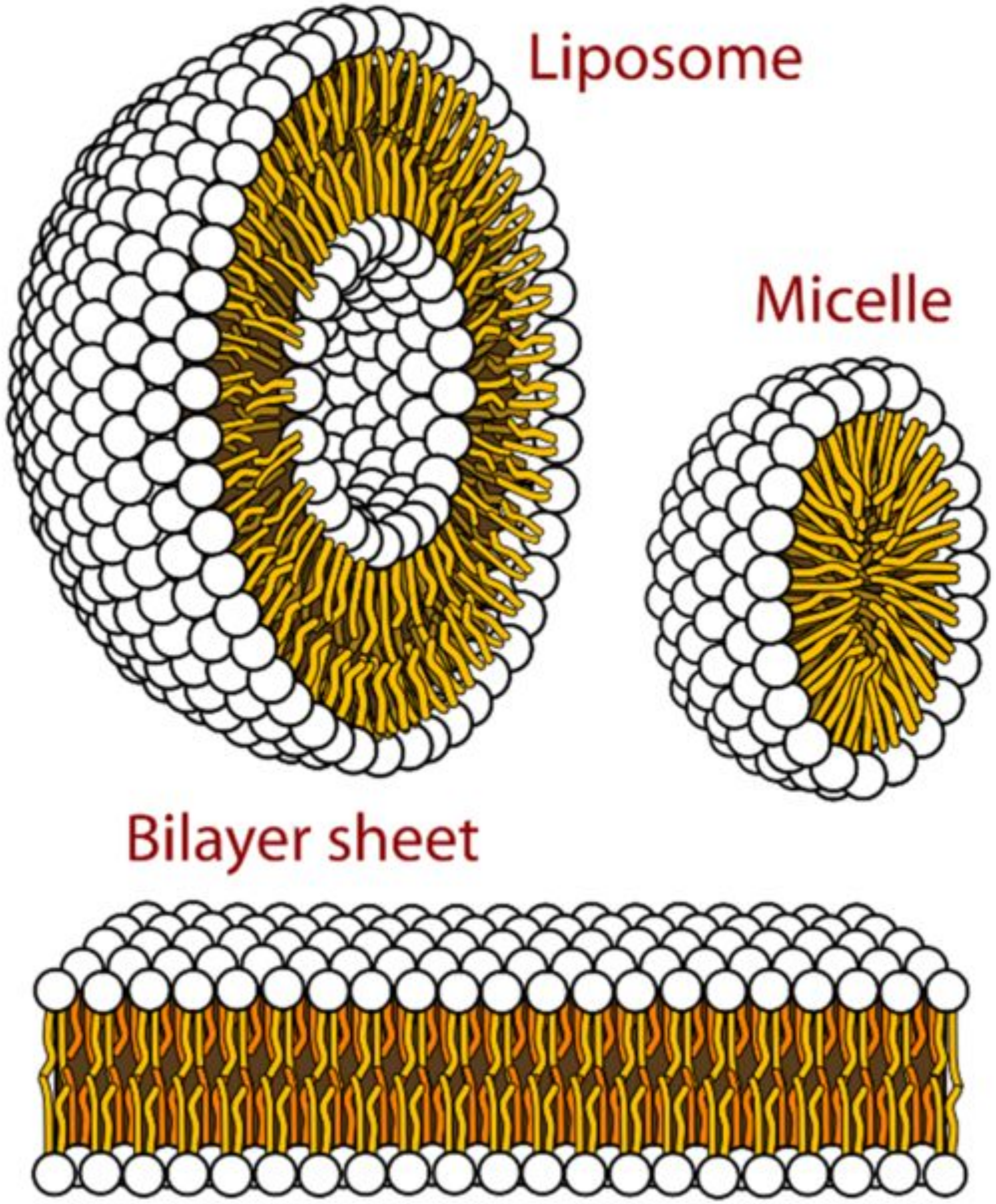


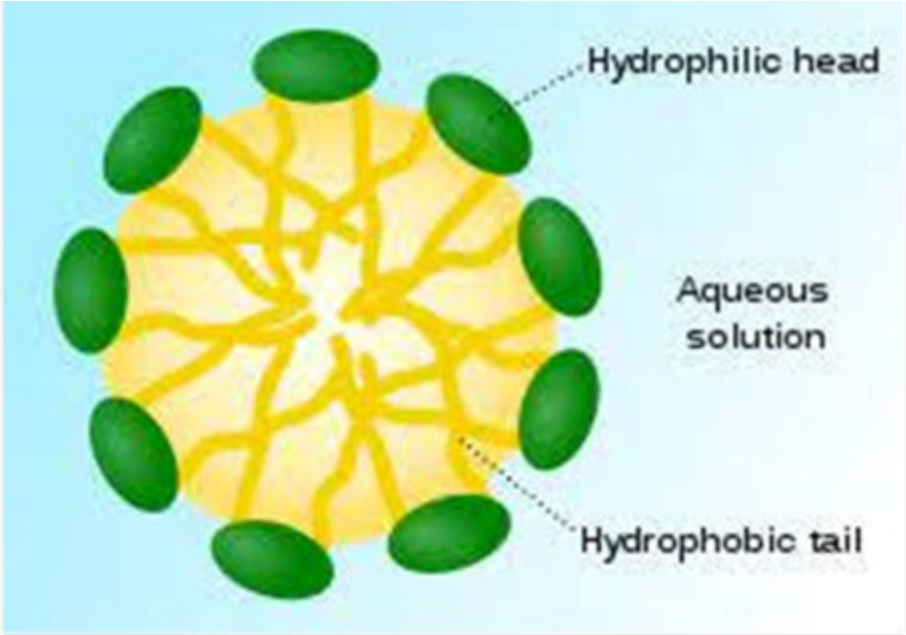


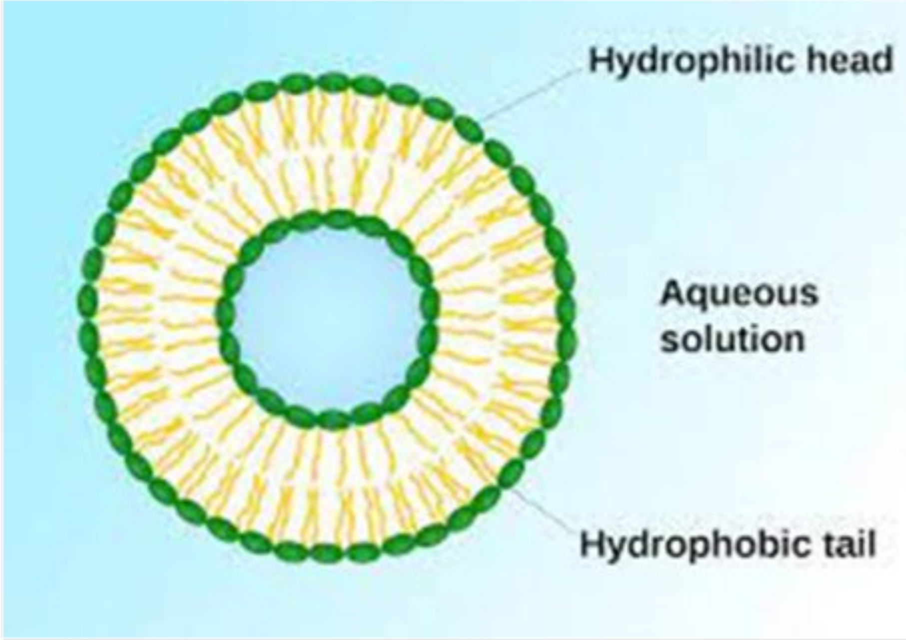


Lipids



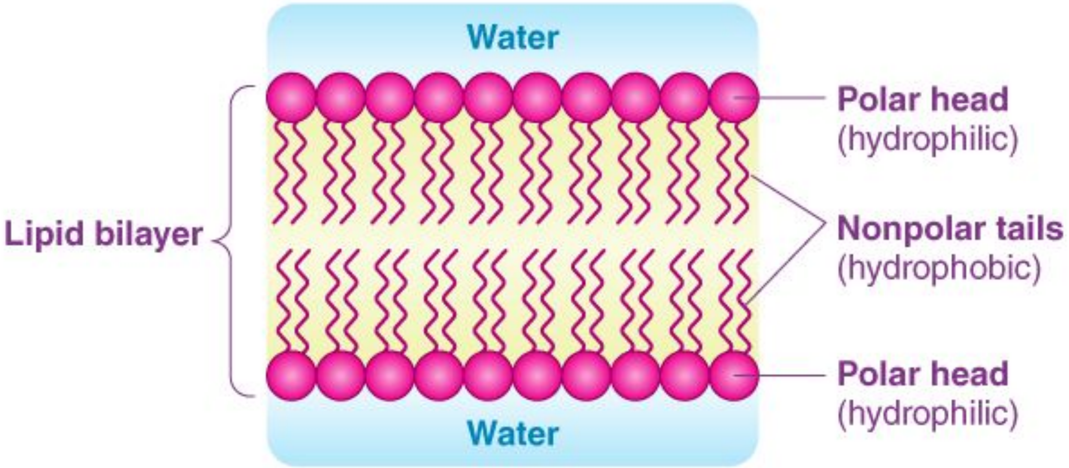

Liposome
Micelle
Bilayer sheet


Hydrophilic head
Aqueous solution
Hydrophobic tail


Hydrophilic head
Aqueous solution
Hydrophobic tail

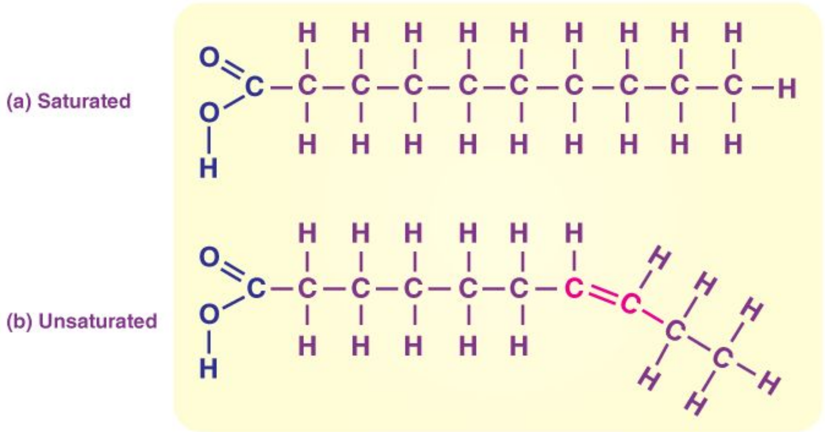
What are Lipids?

These organic compounds are nonpolar molecules, which are soluble only in nonpolar solvents and insoluble in water because water is a polar molecule. In the human body, these molecules can be synthesized in the liver and are found in oil, butter, whole milk, cheese, fried foods and also in some red meats.



Lipid Structure

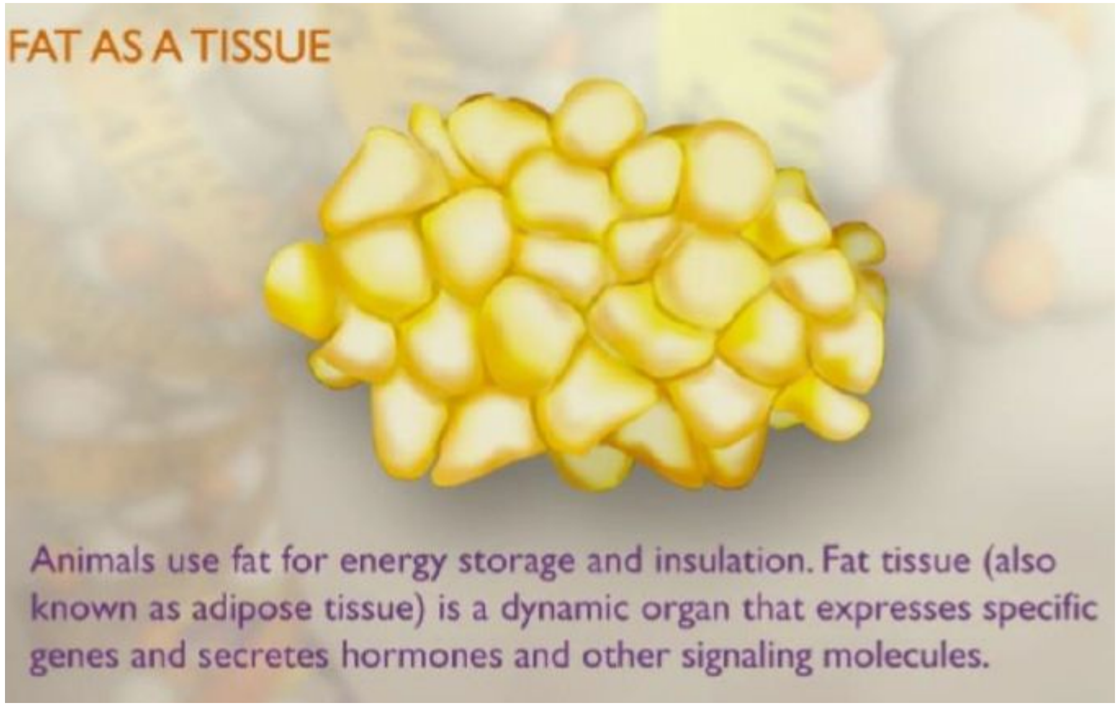
Lipids are the polymers of fatty acids that contain a long, non-polar hydrocarbon chain with a small polar region containing oxygen.

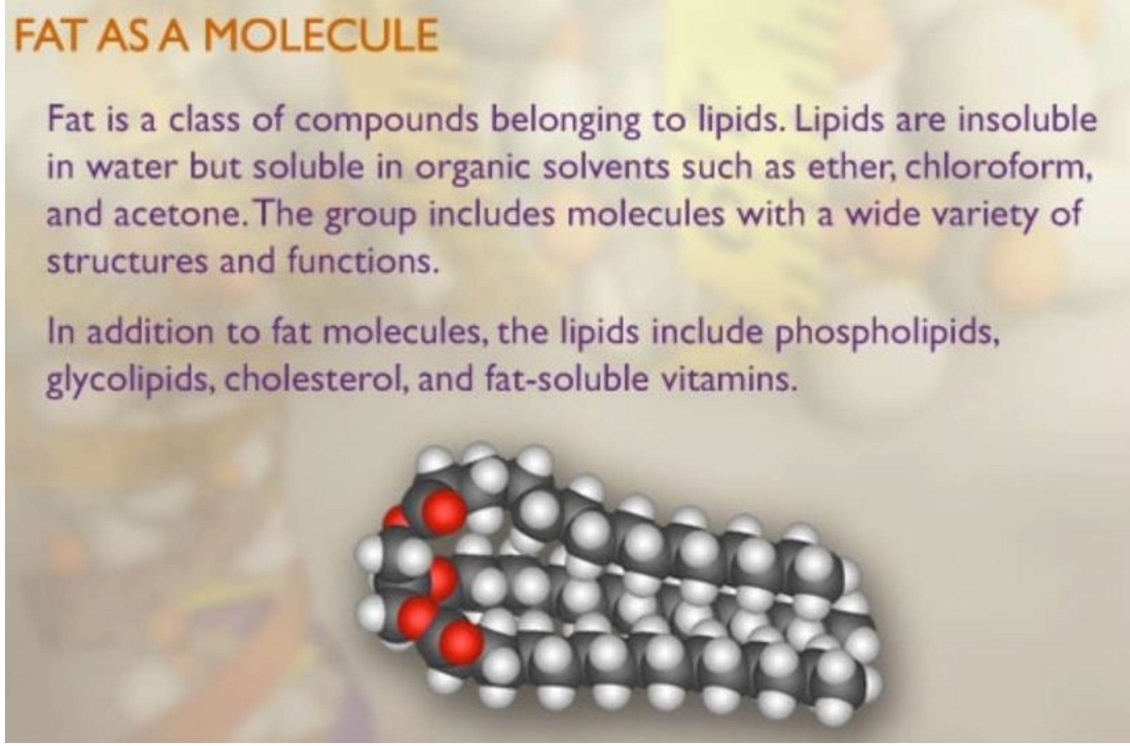


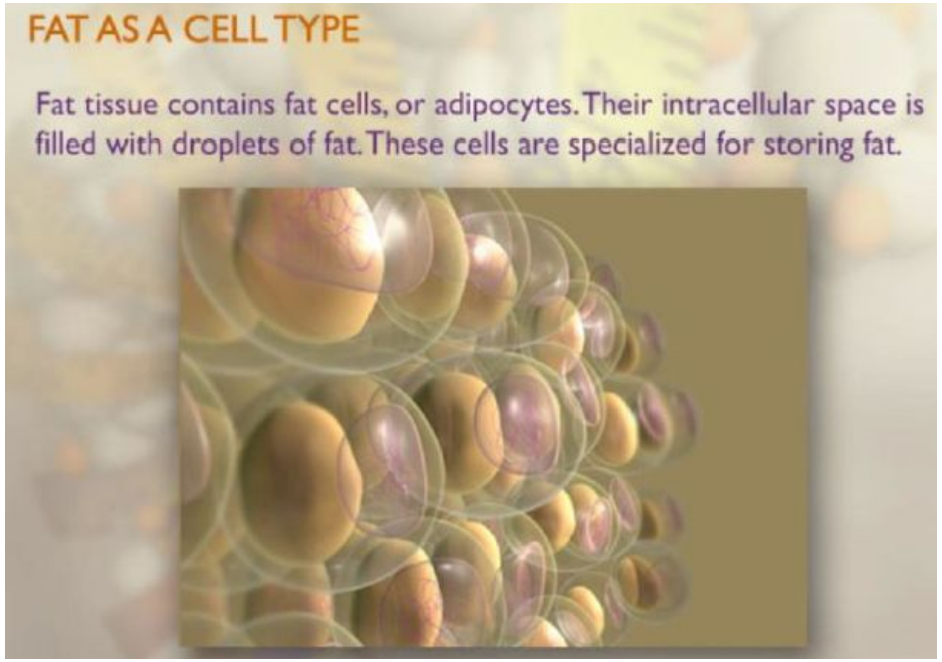
Classification of Lipids

- Nonsaponifiable lipids - A nonsaponifiable lipid cannot be disintegrated into smaller molecules through hydrolysis. Nonsaponifiable lipids include cholesterol, prostaglandins, etc
- Saponifiable lipids - A saponifiable lipid comprises one or more ester groups, enabling it to undergo hydrolysis in the presence of a base, acid, or enzymes, including waxes, triglycerides, sphingolipids and phospholipids.

- Nonpolar lipids, namely triglycerides, are utilized as fuel and to store energy.
- Polar lipids, that could form a barrier with an external water environment, are utilized in membranes.
- Polar lipids comprise sphingolipids and glycerophospholipids.
- Fatty acids are pivotal components of all these lipids.


FAT AS A TISSUE
Animals use fat for energy storage and insulation. Fat tissue (also known as adipose tissue) is a dynamic organ that expresses specific genes and secretes hormones and other signaling molecules.

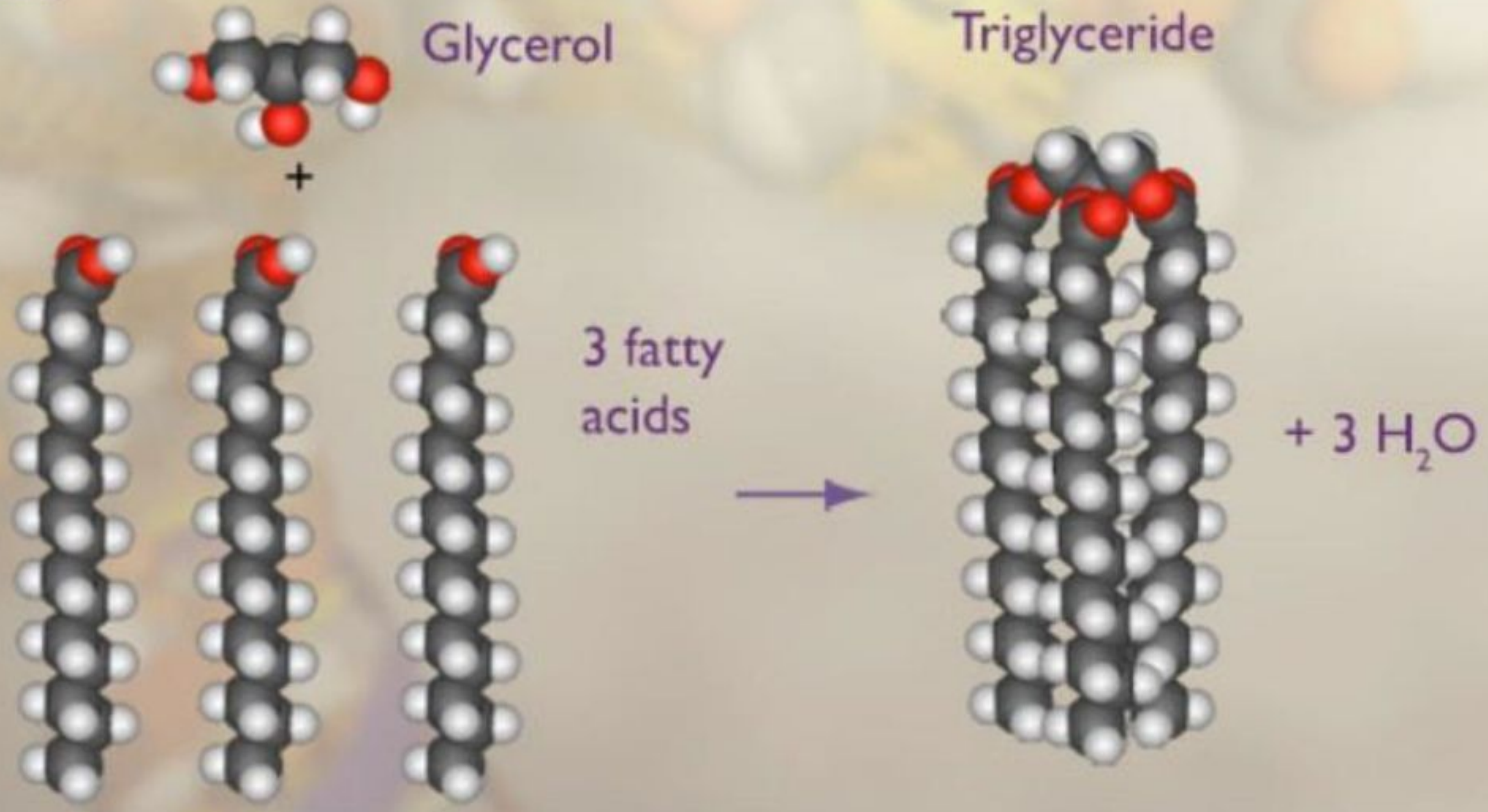

FAT AS A MOLECULE
Fat is a class of compounds belonging to lipids. Lipids are insoluble in water but soluble in organic solvents such as ether, chloroform, and acetone. The group includes molecules with a wide variety of structures and functions.
In addition to fat molecules, the lipids include phospholipids, glycolipids, cholesterol, and fat-soluble vitamins.


FAT AS A CELL TYPE
Fat tissue contains fat cells, or adipocytes. Their intracellular space is filled with droplets of fat. These cells are specialized for storing fat.

The building block

THE FAT MOLECULE

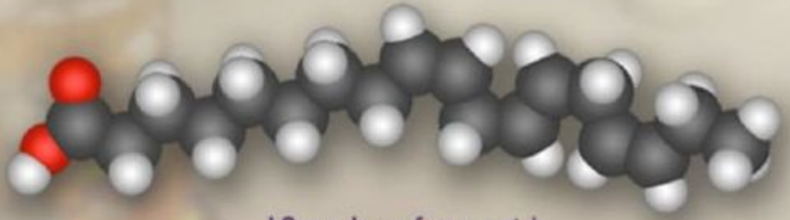
The fat molecule is called a triglyceride. It consists of a glycerol molecule joined to three long carbon-chain organic acids called fatty acids.



Types of fatty acids

FATTY ACIDS

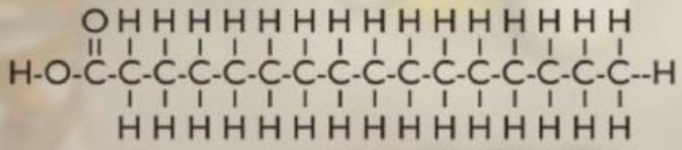
Fatty acids are straight hydrocarbon-chain organic acids. The number of carbon atoms in the chain varies from 4 to about 24. Because fatty acids are synthesized by extending the chain two carbon atoms at a time, the number of carbon atoms in the chain is usually an even number. In higher organisms, 16 and 18 are the most common.

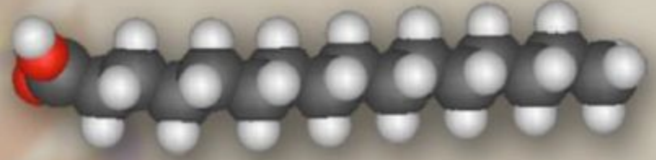


18-carbon fatty acid

SATURATED FATTY ACIDS

Fatty acids may be saturated. This means that no more hydrogen atoms can be added to the molecule. All bonds between carbon atoms are single bonds. The hydrocarbon chain is straight in saturated fatty acids. Triglycerides with three saturated fatty acids are called saturated fats.

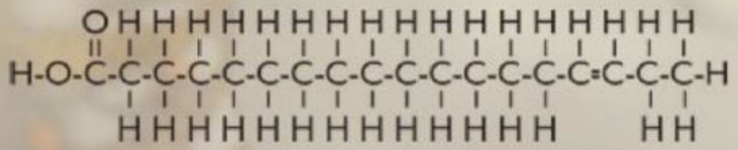




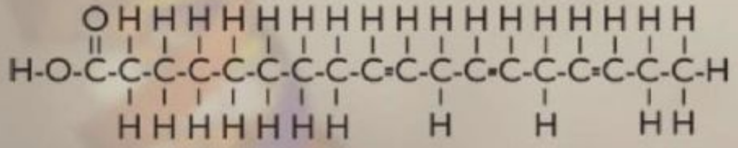
Palmitic acid (16-carbon saturated fatty acid)

UNSATURATED FATTY ACIDS

Some fatty acids have one double bond or more between the carbon atoms. They are called unsaturated fatty acids because they could hold more hydrogen atoms. They can be monounsaturated (one carbon-carbon double bond) or polyunsaturated (multiple carbon-carbon double bonds). Triglycerides with at least one unsaturated fatty acid are called unsaturated fats.

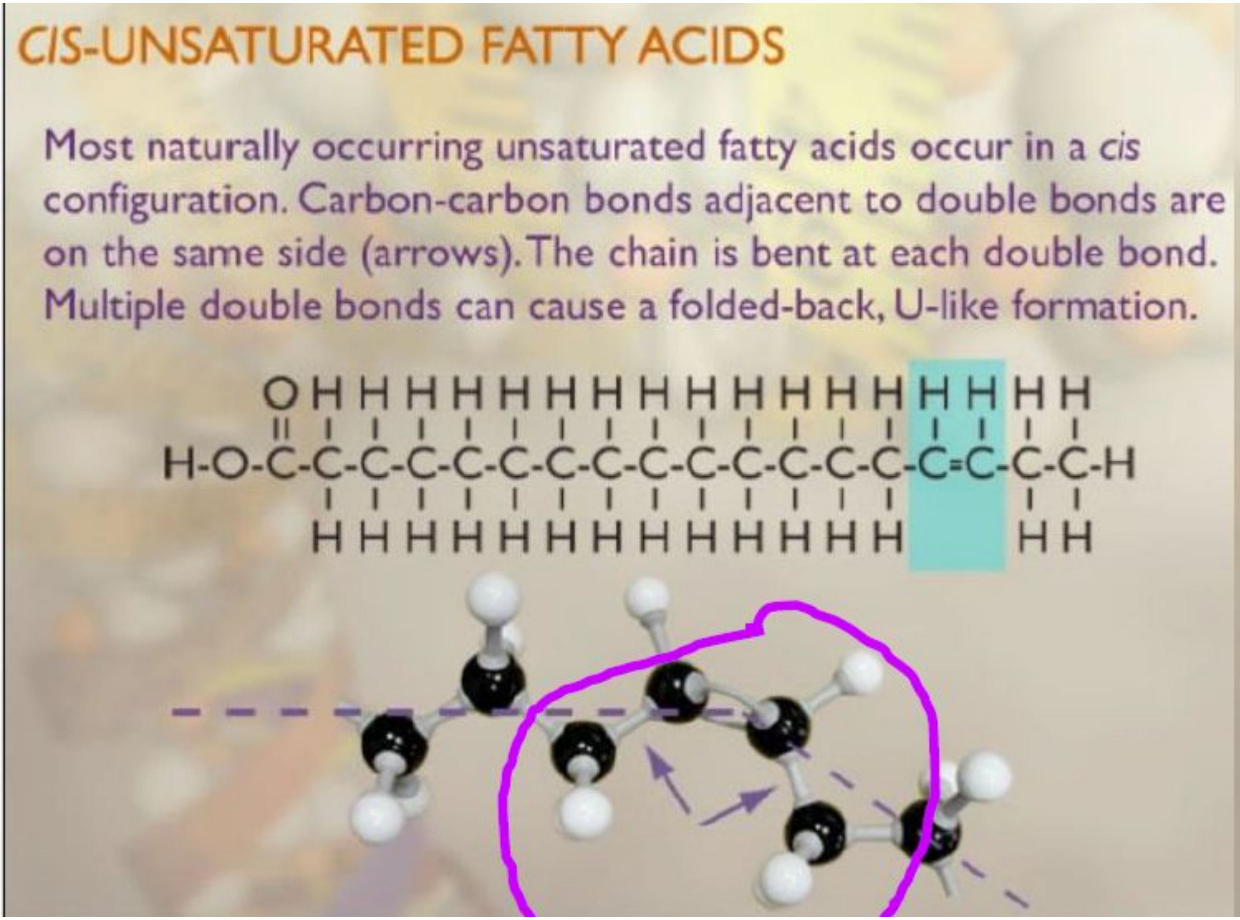


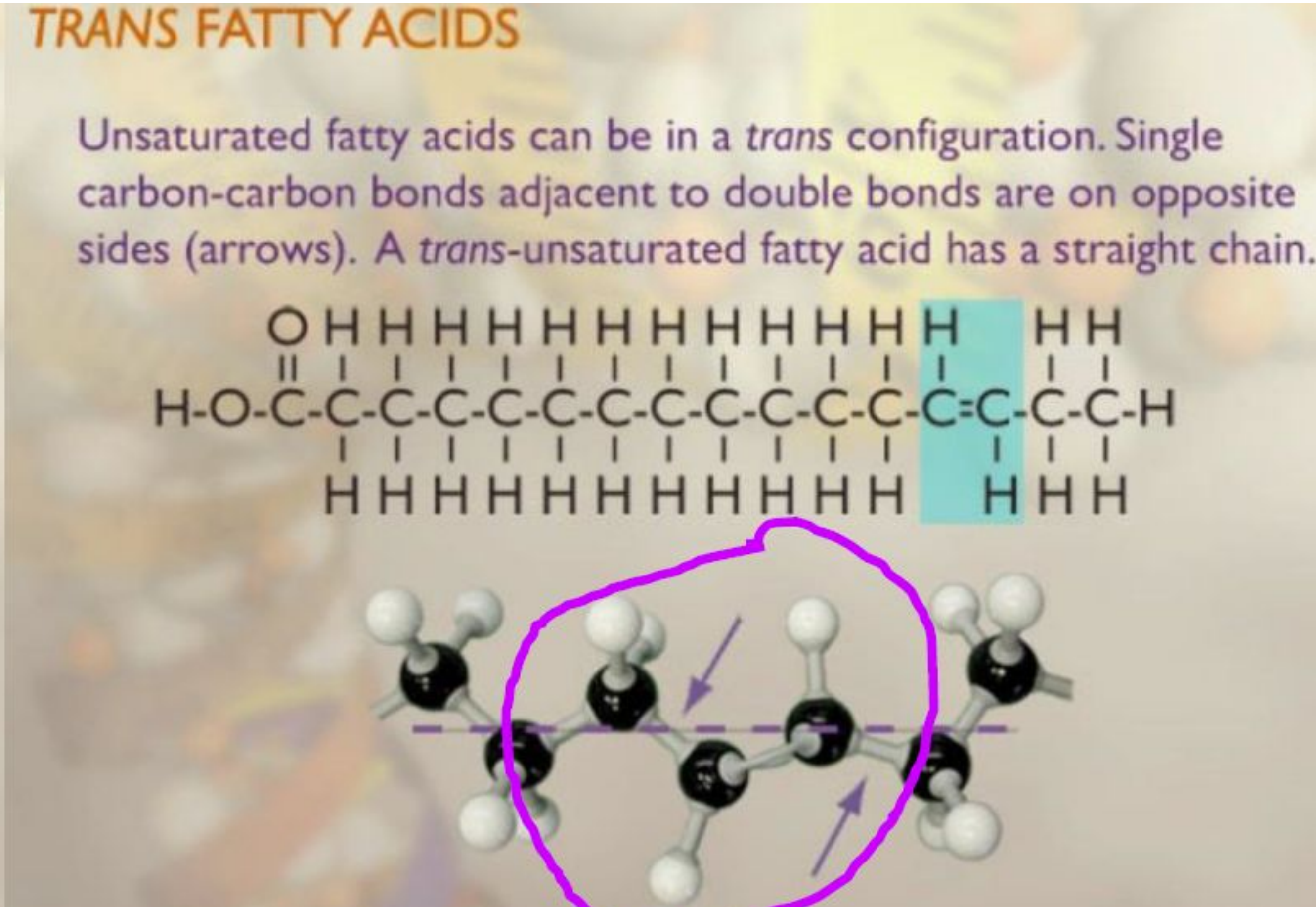
Monounsaturated



Polyunsaturated

Types of unsaturated fatty acids



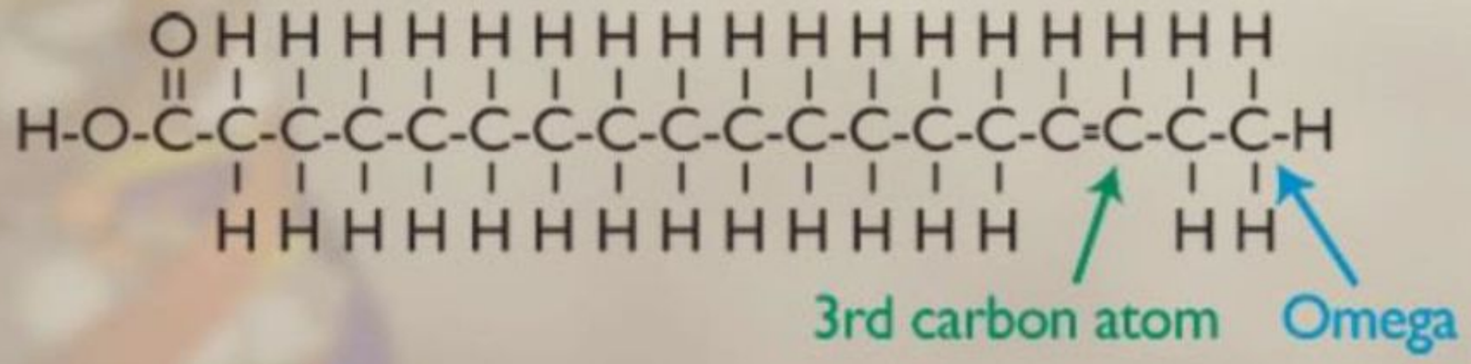


The good fatty acids

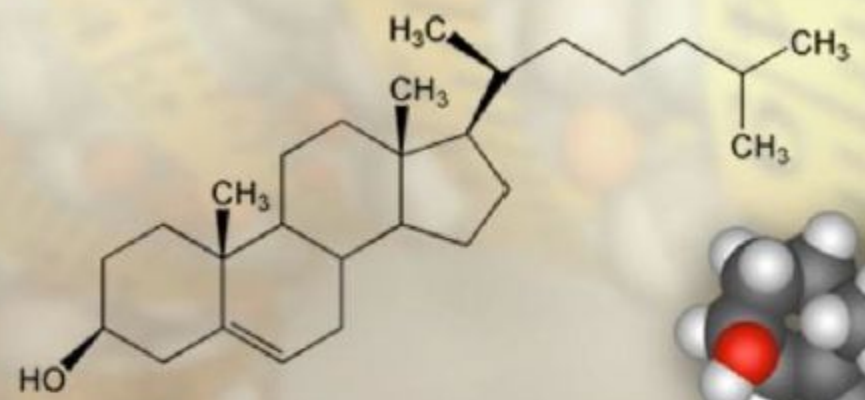
OMEGA-3 FATTY ACIDS

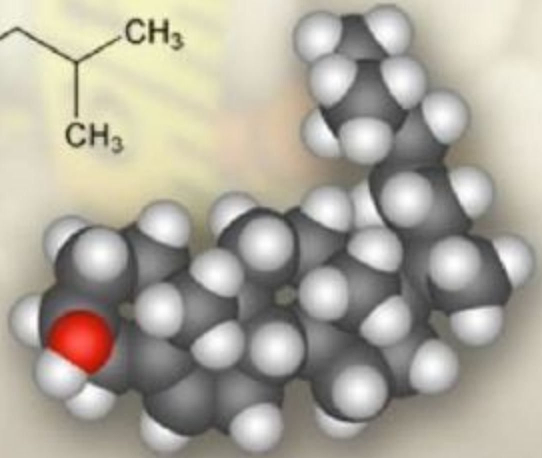
"Omega-3" is a term from a naming system for both mono- and polyunsaturated fatty acids. "Omega" refers to the last carbon atom away from the carboxyl group in the fatty acid chain. "Omega-3" means there is a double bond between the third and the fourth carbon atoms counting from the omega carbon atom.

Naturally-occurring omega-3 fatty acids are thought to help protect against cardiovascular disease.



OTHER LIPIDS: CHOLESTEROL

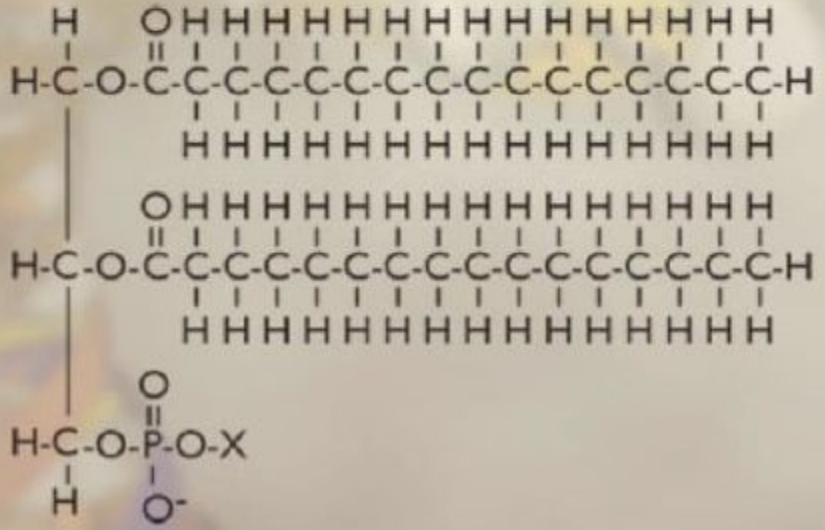




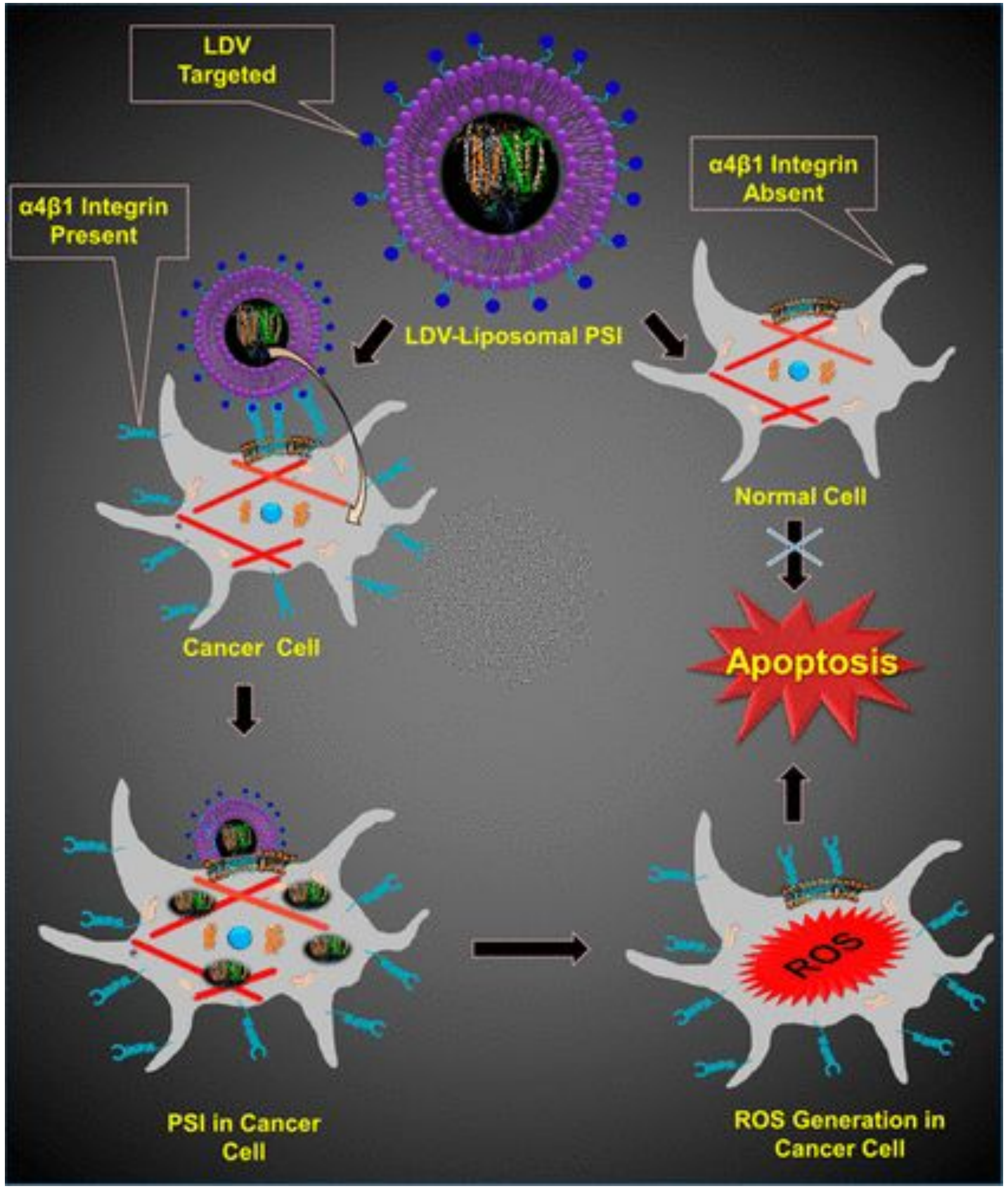
Cholesterol is a type of lipid that is different from fat. An important molecule in the cell membrane, it is also a precursor to various other key molecules, including bile acids and steroid hormones.

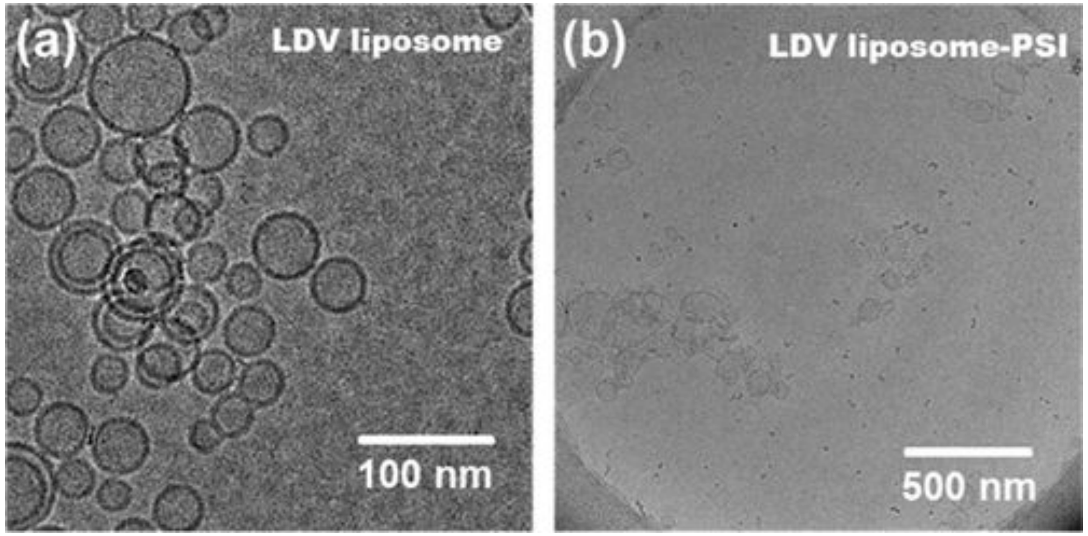
OTHER LIPIDS: PHOSPHOLIPID

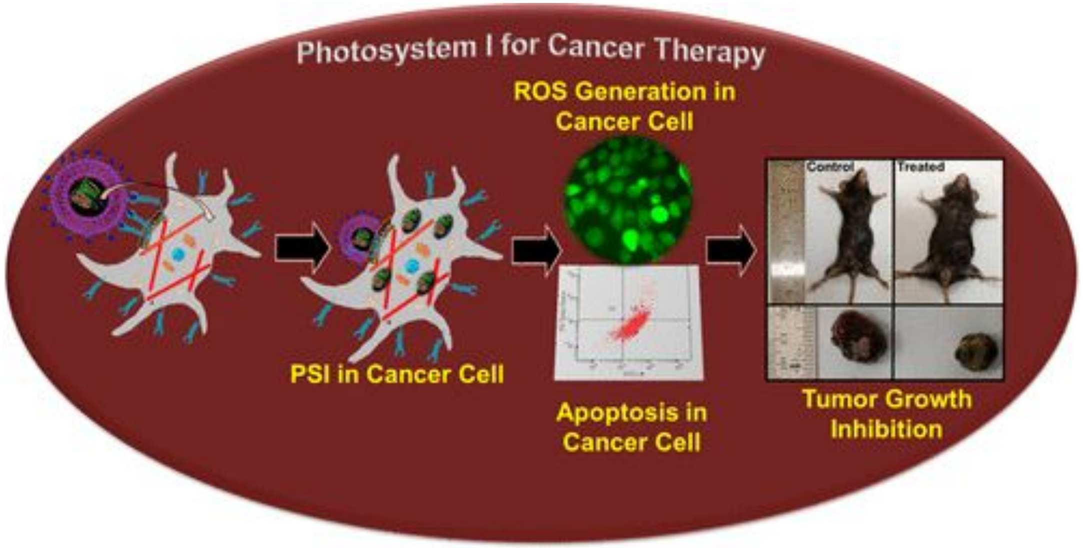
Phospholipids (or phosphoglycerides) are the main molecules that make up the lipid bilayer in biological membranes. They have two fatty acid chains and a phosphate group with a variable group X.



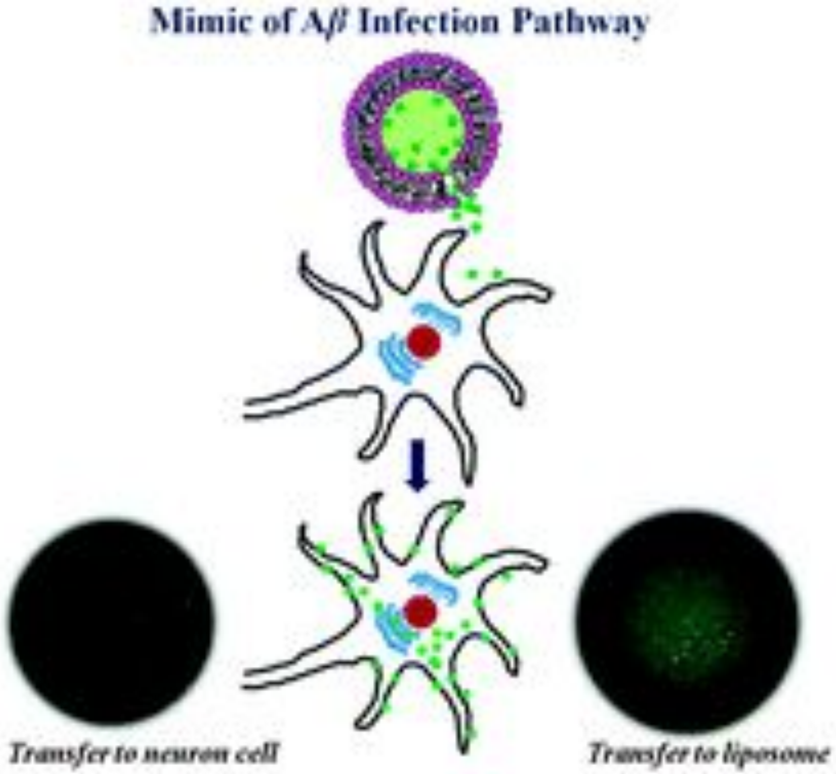
Liposomes as drug delivery agents

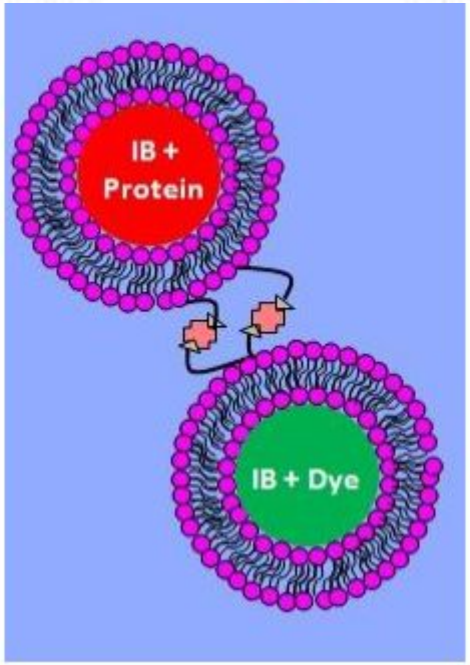


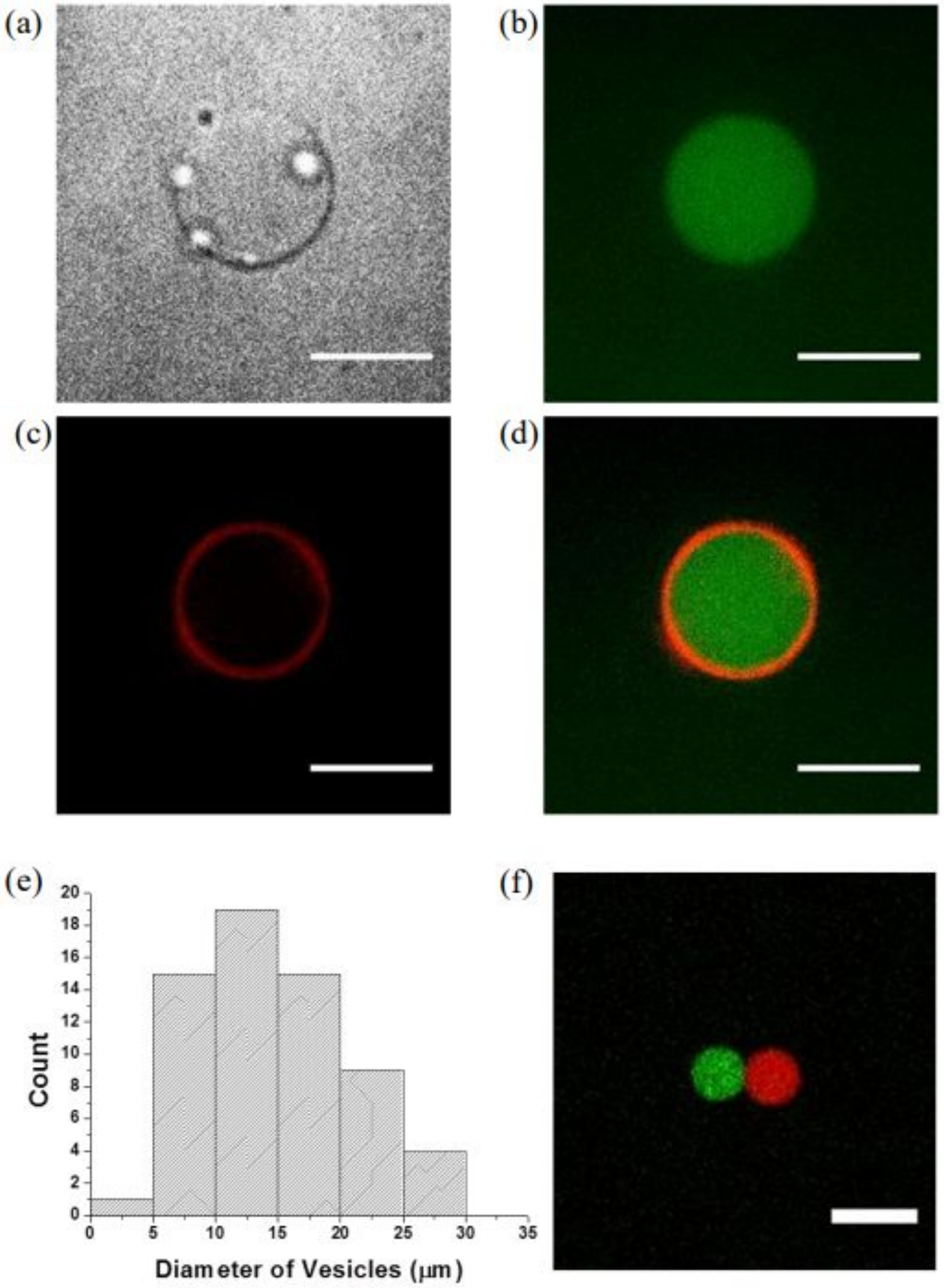




Saha A., Ghosh S., et al, 2017







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All macromolecules together